Dell PowerFlex 4.5.x and PowerFlex Manager 5.1.x

Technical Overview

October 2026
Rev. 1.1

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Introduction

This document provides a high-level technical overview of PowerFlex. It describes the key PowerFlex features and the management capabilities of the PowerFlex management platform. This document also contains information regarding the different PowerFlex offerings.

The target audience for this document includes customers, presales, and Dell Technologies Support.

Integrated PowerFlex management (non-KVM) was previously known as co-resident. Customer-provided infrastructure for PowerFlex management (Linux/KVM) was previously known as bring your own Linux. References in the documentation will be updated over time.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
management virtual machine	MVM
management data store	MDS
metadata manager	MDM
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
storage data target	SDT
system configuration reporter	SCR

The following table summarizes key terminology changes. References in the documentation will be updated over time.

Table 2. Terminology changes

Term	Formerly known as
PowerFlex management cluster (ESXi)	PowerFlex management controller 2.0
PowerFlex management cluster (vSAN)	PowerFlex management controller 1.0
PowerFlex management cluster	PowerFlex management controller
Single ESXi PowerFlex management node	PowerFlex management node
PowerFlex management node	PowerFlex controller node

Table 2. Terminology changes (continued)

Term	Formerly known as
Dell Technologies connectivity services	SupportAssist
Enterprise Project Services (EPS) form	Logical Configuration Survey (LCS)
Dell Code and Compatibility Knowledge Hub	Release Certification Matrix (RCM) Portal

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 3. Revisions

Date	Document revision	Description of changes
October 2026	1.1	Editorial updates
June 2026	1.0	Initial release

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PowerFlex overview

PowerFlex is an enterprise-class, software-defined block storage solution that is deployed, managed, and supported as a single system. Highly flexible and scalable, PowerFlex allows customers to have a mixture of two-layer (server SAN), single-layer hyperconverged configurations, storage-only, and other mixed architectures within a single deployment.

With PowerFlex, resources such as storage and compute can be scaled together or separately, nondisruptively, and in small increments. The system can scale from a few nodes to hundreds in a cluster, linearly scaling I/O performance and throughput. In addition, PowerFlex supports a broad set of operating environments that include multiple hypervisors, bare-metal operating systems, and container management tools. Using native asynchronous replication, customers have the flexibility to replicate data across multiple data centers or to cloud providers such as AWS and Azure. This allows customers to transition application architectures, while supporting multiple generations of applications with disparate architectures in a multicloud environment.

PowerFlex offers integrated product options. PowerFlex appliance is a scalable system with flexible form factors that comes preconfigured and validated for fast, easy deployment. PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support. PowerFlex custom node is a validated engineering solution that provides nearly all the flexibility of the PowerFlex software offering.

PowerFlex also supports multicloud deployments through its Dell PowerFlex for AWS and Dell PowerFlex for Microsoft Azure offerings.

Terminology for management offerings

The following table describes important PowerFlex Manager terms and concepts for the PowerFlex management offerings.

Table 4. Terms and concepts

Term	Description
Customer-provided infrastructure for PowerFlex management (KVM or ESXi or Linux)	Allows customer to supply infrastructure to host the Kubernetes VM or environment. Enables setup of the PowerFlex management platform system for management purposes in VMware ESXi or Kubernetes VM.
Integrated PowerFlex management (non-KVM)	The PowerFlex storage-only nodes in a protection domain runs the management of the system.
Management data store (MDS)	The PowerFlex management data store is on the dedicated multi-node controller. It is not the same as the data store used in an integrated PowerFlex management (non-KVM) option. Kubernetes VMs host the management containers that reside on this data store.
Integrated PowerFlex management (KVM)	A KVM solution that manages a minimum of three storage nodes for PowerFlex appliance. This solution can provide deployment on a minimum of three storage nodes where PowerFlex Manager and PowerFlex storage co-reside on the same node.
Management virtual machine (MVM)	The PowerFlex Manager UI is hosted in a Kubernetes environment running across a series of Linux machines. The VM version is called the management VM. The operating system is a customized SLES. version maintained by Dell Technologies.
PowerFlex management cluster (KVM)	PowerFlex management is enabled with a Linux operating system. This new management cluster is based on a KVM solution to host management virtual machines. This solution can provide deployment on three dedicated management nodes hosting multiple instances of PowerFlex management platform for PowerFlex appliance and PowerFlex rack.

Table 4. Terms and concepts (continued)

Term	Description
PowerFlex management cluster (ESXi)	The infrastructure underlying the PowerFlex management platform in a dedicated multi-node configuration. It is mandatory in PowerFlex rack, and optional for PowerFlex appliance. The physical controllers with VMware ESXi hosting PowerFlex management platform can be a three-node or a five-node VMware ESXi cluster with: • PowerFlex as shared storage (PowerFlex management cluster (ESXi)) • vSAN (PowerFlex management cluster (vSAN)) Use in upgrades from PowerFlex 3.8.x systems only. PowerFlex management cluster also hosts VMs for vCenter, a jump server, and other optional NSX or CloudLink managers.
PowerFlex management platform	The management software stack for PowerFlex Manager runs on physical or virtual Linux instances, which are containers that provide services. This management solution can be implemented on various infrastructure options.

New features for mirroring

This PowerFlex release introduces the following new mirroring features:

Table 5. New features

Feature	Details
CloudLink	Added support for CloudLink 8.3. CloudLink 8.3 and later versions are supported only on PowerFlex platforms. **NOTE:** CloudLink 8.2 The Key Management Interoperability Protocol (KMIP) server and machine encryption functionalities are available. CloudLink 8.3 KMIP server and machine encryption functionalities remain operational but are deprecated. They are no longer covered under standard support and are planned to be disabled in a future release. Dell Technologies strongly recommends planning a transition to supported alternative solutions to ensure continued functionality and support. For guidance on available options or migration planning, contact your Dell Technologies account representative.
Dynamic licensing	In this release, PowerFlex supports dynamic licensing. Dynamic Licensing offers an enhanced licensing experience by automating license management activities, such as registrations, capacity expansions, and subscription renewals. You can register software instances and share licensing information between the product instance and Dell. Dynamic licensing introduces the ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. Through

Table 5. New features (continued)

Feature	Details
	PowerFlex Manager, you can register your device and migrate from key-based XML format licensing to dynamic licensing. See the *Dell PowerFlex Family Licensing Guide* for more information.
Pause inventory jobs	In this release, PowerFlex Manager introduces the option to pause scheduled inventory jobs for up to 24 hours, ensuring upgrades can complete without interruption.
User experience enhancements	In this release, PowerFlex Manager delivers a simplified user experience that is related to storage pools, volumes, and snapshots and enhanced capacity metrics reporting. ● You can view the security and expiration that is associated with a volume snapshot. ● The following capacity metrics are now available through the PowerFlex Manager dashboard: ○ Reducible DRR: The data reduction ratio for the reducible data within the system or protection domain. This metric ignores any unreducible data in the calculation. ○ Unreducible data: The amount of data that is written within the system or protection domain which cannot be reduced by compression or the pattern detection logic PowerFlex uses.

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System requirements and product limits

The following section describes the PowerFlex product limits, and the system requirements for components for each consumption model. The product limits summarize the supported capabilities and system limits.

Maximum and minimum values for key system components are dictated by the product limits.

Product limits

The following table summarizes PowerFlex supported capabilities and system limits.

NOTE: The supported limits that are described in the following table might differ from system-enforced limits that are listed in the output for the `scli --query_system_limits` command.

Table 6. Supported capabilities and system limits

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
Device size	15.36 TB maximum on medium granularity storage pool 7.86 TB maximum on fine granularity storage pool	15.36 TB maximum on medium granularity storage pools 7.86 TB maximum on fine granularity storage pools
Maximum acceleration devices of type SDPM or NVDIMM (maximum that is supported in a PowerFlex node)	PowerEdge R840/R740xd/R640 nodes: • NVDIMM: 6 PowerEdge R750/R650 nodes: • NVDIMM: 8 PowerEdge R760/R660 nodes: • SDPM: 2	PowerEdge R840/R740xd/R640 nodes: • NVDIMM: 6 PowerEdge R750/R650 nodes: • NVDIMM: 8 PowerEdge R760/R660 nodes: • SDPM: 2
Maximum data networks for MDM virtual IP v4 addresses	4	4
Maximum data networks in the system	8 **NOTE:** SDCs can connect through a router and exceed this limit.	4 **NOTE:** SDCs can connect through a router and exceed this limit.
Maximum devices per protection domain	8,192	768
Maximum devices (drives) per SDS server **NOTE:** Includes SDPM or NVDIMM devices.	64 On a VMware ESXi server, the maximum devices per SDS are 59.	32
Maximum devices per storage pool	300 If more are needed, contact Dell Technologies Support.	300 If more are needed, contact Dell Technologies Support.
Maximum fault sets per protection domain	64	64

Table 6. Supported capabilities and system limits (continued)

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
Maximum fine granularity compression ratio	10x the system's raw capacity	10x the system's raw capacity
Maximum logical capacity size of all volumes per storage pool	4 PB	4 PB
Maximum logical capacity size per system	16 PiB	16 PiB
Maximum managed PowerFlex nodes per system	Not applicable	773
Maximum number of concurrent logged-in management clients (UI/ REST/CLI)	128	128
Maximum number of configured syslog servers	16, must be an even number	16, must be an even number
Maximum number of snapshots a snapshot policy can be defined to retain per vTree (not including locked snapshots)	60	60
Maximum number of snapshots in a single vTree	126	126
Maximum number of volumes to SDC mappings per system	262,143	262,143
Maximum PowerFlex nodes per resource group	32	32
Maximum protection domains per system	256	256
Maximum raw capacity per protection domain	8 PB	8 PB
Maximum raw capacity per SDS	160 TB (medium granularity) 128 TB (fine granularity)	160 TB (medium granularity) 128 TB (fine granularity)
Maximum resource groups per system	63	63
Maximum snapshot policies per system	1,000	1,000
Maximum SDCs per system	2,048	768 managed nodes per system, including all node types, Unmanaged hosts align to 2,048.
Maximum SDSs per protection domain	128	128
Maximum SDSs per system	512	400 storage-only or VMware ESXi - based PowerFlex hyperconverged nodes per system or both. The total number of managed nodes (storage-only, VMware ESXi - based PowerFlex hyperconverged, and PowerFlex compute-only) cannot exceed 400.
Maximum storage pools per protection domain	64	10
Maximum storage pools per system	1,024	1,024

Table 6. Supported capabilities and system limits (continued)

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
Maximum system raw capacity	Up to 16 PB	Up to 16 PB
Maximum total number of volumes and snapshots in protection domain	32,768	32,768
Maximum total number of volumes and snapshots in the system	131,072 Up to 10,000 volumes and snapshots when deploying NVMe over TCP.	131,072
Maximum total number of volumes and snapshots per storage pool	32,768	32,768
Maximum user accounts	256	256
Maximum volumes that can be mapped to a single SDC	1,024	1,024
Maximum volumes that can be mapped using API (scli, REST, or web UI) concurrently	1,024	1,024
Minimum devices (drives) per storage pool	3, one per fault unit	For PowerFlex appliance: 3, one per fault unit For PowerFlex rack: 4, one per fault unit
Minimum number of SDSs in a storage pool	3	For PowerFlex appliance: 3 For PowerFlex rack: 4
Minimum storage pool size	720 GB	720 GB
RMcache	Minimum: 128 MB Maximum: 300 GB	Not applicable
SCSI-3 reservation type	Write exclusive registrants only	Write exclusive registrants only
System over-provisioning factor	5x net capacity per medium granularity layout Maximum 16 PiB provisioned volumes	5x net capacity per medium granularity layout Maximum 16 PiB provisioned volumes
Volumes per local consistency group (snapshot)	1,024	1,024
Volume size	Minimum: 8 GB Maximum: 1 PB	Minimum: 8 GB Maximum: 1 PB

Table 7. NVMe over TCP limits

Item	Limit
Maximum volumes mapped to a single NVMe host (Linux)	1024 VMware ESXi 8.0 and VMware ESXi 7.0u3f or higher
Maximum volumes mapped to a single NVMe host (VMware ESXi)	256 using VMware ESXi 8.0 32 with VMware ESXi 7.0u3f
Maximum NVMe hosts connected to system	1024

Table 7. NVMe over TCP limits (continued)

Item	Limit
	NOTE: These hosts are in the total number of SDCs per system.
Maximum SDTs per protection domain	128
Minimum SDTs per protection domain	2 Using minimum SDTs may block the ability to reach maximum NVMe hosts.
Maximum SDTs per system	512
Maximum paths in multipathing driver per volume	8 4 in VMware ESXi 7.0u3f
Maximum connections per host per protection domain	16
Maximum NVMe host connections (I/O controllers) per SDT	512
Maximum NVMe host connections (I/O controllers) per system	65,519
Maximum I/O controller queue depth	128
Maximum I/O controller queues	32 **NOTE:** Number of queues + queue depth is automatically negotiated on the connection.
Maximum volume-to-host mappings (SDC/NVMe) per system	262,143

NOTE: Replication is not supported on NVMe over TCP-based volumes.

Table 8. Replication limits

Item	Limit
Number of destination systems for replication	4
Maximum SDRs per system	128
Maximum number of replication consistency groups	1,024
Maximum replication pairs in replication consistency group with initial copy	1,024
Maximum number of volume pairs per replication consistency group	1,024
Maximum volume pairs per system	32,000
Maximum number of remote protection domains	6
Maximum number of copies per replication consistency group	1
Maximum ratio of peer SDRs in a source protection domain to a target protection domain	1:6
Maximum ratio of maintenance SDRs in a source protection domain to a target protection domain	1:18
Recovery point objective (RPO)	Minimum: 15 seconds Maximum: 1 hour
Maximum replicated volume size	64 TB

Table 9. PowerFlex management platform limits

PowerFlex management platform capability	Limit
Maximum number of configured users	20
Maximum number of LDAP users	20
Maximum number of LDAP servers	1
Maximum number of DNS servers	1
Maximum number of time sources	1
Maximum number of syslog servers	1
Maximum number of SMTP servers	1
Maximum number of vCenters	1

Table 10. Deployment and NDU limits

PowerFlex item	Limit
Maximum latency for MDM network	200 msec
SDS IP addresses	Must be unique. For example, 127.0.0.1 must not be used, as it refers to several machines.

Related information

PowerFlex resource requirements
SAN virtualization layer

System requirements

This section lists the requirements for PowerFlex system components and the minimum PowerFlex software resource requirements for each consumption model.

NOTE: PowerFlex software does not contain the VMware ESXi and vCSA images and the patch depot file. Customers must procure and provide the VMware ESXi and vCSA images and patch depot file for building the PowerFlex management cluster, PowerFlex hyperconverged nodes, or PowerFlex compute-only nodes from the Broadcom support portal.

Required software packages

PowerFlex 4.5.x and PowerFlex Manager 4.8.x deployments require the following software packages:

Table 11. Software packages

PowerFlex consumption model	Package
All PowerFlex models	PowerFlex software ZIP
PowerFlex custom node	Custom node software ZIP

For more information about software package names and versions, see the *PowerFlex and PowerFlex Manager Release Notes*.

PowerFlex resource requirements

The following section lists the requirements for PowerFlex system components and the minimum PowerFlex management platform resource requirements.

The following table summarizes the requirements for node count, volumes, and SDS devices based on the cluster scale of the deployment.

Table 12. PowerFlex requirements based on cluster scale

System scale	Node count	Volumes	SDS devices in the system
Low scale	From 3 to 10	From 0 to 1,000	From 3 to 240
Medium scale	From 11 to 50	From 1,000 to 10,000	From 241 to 1,200
High scale	From 51 to the maximum limit See *Product Limits* for the nodes maximum limit.	From 10,000 to the maximum limit See *Product Limits* for the volume maximum limit.	From 1,201 to the maximum limit See *Product Limits* for the SDS devices maximum limit.

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex software, depending on the cluster scale.

Table 13. PowerFlex software storage and resource requirements

System scale	CPU cores	Storage capacity (GB)	RAM (GB)
Low scale	8	300	24
Medium scale	8	350	26
High scale	14	450	32

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex rack and PowerFlex appliance.

Table 14. PowerFlex rack and PowerFlex appliance storage and resource requirements

CPU cores	Storage capacity (GB)	RAM (GB)
14	600	32

Related information

Product limits

PowerFlex cluster component requirements

Each PowerFlex cluster includes the following components. Before deploying PowerFlex, ensure that the PowerFlex environment meets the server requirements for the cluster type.

Three-node cluster

- One primary MDM
- One secondary MDM
- One tiebreaker
- Minimum of three SDSs (on the same servers as the above components, or on three different servers)
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

Five-node cluster

- One primary MDM
- Two secondary MDMs
- Two tiebreakers
- Minimum of three SDSs (on the same servers as the above components, or on three different servers) Dell Technologies recommends five SDSs so that the cluster members reside with the SDSs.
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

PowerFlex management platform options

- PowerFlex appliance only: Integrated PowerFlex management (KVM).
- PowerFlex rack and PowerFlex appliance: PowerFlex management cluster (KVM).
- PowerFlex custom node: One node, dedicated PowerFlex management cluster.

  NOTE: In the one-node configuration, a hypervisor is required, such as VMware ESXi, to host PowerFlex management platform VMs.

- PowerFlex custom node or PowerFlex software: Integrated PowerFlex management (non-KVM)—available for PowerFlex software and custom node offerings.
- PowerFlex software or PowerFlex custom node offerings: Customer-provided infrastructure for PowerFlex management (ESXi) to host the PowerFlex management cluster.

Physical server requirements

The following section lists the physical server requirements for PowerFlex and recommended best practices.

The recommended best practice when installing new nodes is to keep nodes with similar hardware characteristics in their own protection domain. Different CPUs, memory speeds, drive capacity, drive types, and network speeds can affect the overall node performance. PowerFlex Manager does not prevent the mixing of node types within a protection domain. The lowest performing system dictates performance, which might not always be the preferred option.

PowerFlex offers a sizing tool to make the complex work of sizing the physical requirement. Go to the PowerSizer and review the physical sizing considerations on the page. Dell Technologies recommends using the sizer tool for all and not perform manual sizing.

This table summarizes the physical server requirements for Dell PowerEdge servers (R750/650).

NOTE: Physical memory requirements for Dell PowerEdge servers (R760/R660) include non-volatile random-access memory (SDPM), Dell PowerEdge servers (R750/R650) include non-volatile random-access memory (NVDIMM).

Table 15. Physical server requirements

Component	Requirement
Processor	One of the following: • Intel or AMD x86 64-bit (recommended) • Intel or AMD x86 32-bit (for Citrix Hypervisor (Xen) only) **NOTE:** • Only current customers who have installed XenServer or Citrix Hypervisor are supported, new customers are not advised to install XenServer or Citrix Hypervisor. • AMD processors are not supported on VxFlex Ready Node and do not support SDPMs or NVDIMMs. There is no support for fine granularity storage pools on AMD-based servers. CPU threads: • MDM: 5 • SDS: Default 8/maximum 12 • CloudLink: 2 • LIA: 1 • SDR: 8

Table 15. Physical server requirements (continued)

Component	Requirement
	• NVMe target: 8 • SDC: Default 2/maximum 8 • NVMe initiator: Default 2/max 8 **NOTE:** The recommended best practice is to have a different non-uniform memory access (NUMA) domain than SDR if it is on the same operating system.

Physical memory	PowerFlex component requirements:
	RAM (GiB)
	• SDS FG: 10 GiB + ((100*Number_Of_Drives) + (550 * Total_Drive_Capacity_in_TiB))/1024
	• SDS MG: 5 GiB + (210*Total_Drive_Capacity_in_TiB)/1024
	• SDS SDPM or NVDIMM allocation for FG: SDPM_capacity_in_GiB or NVDIMM_capacity_in_GiB = ((100*Number_Of_Drives) + (700 * Total_Drive_Capacity_in_TiB))/1024
	• MDM: 6.5 GiB
	• CloudLink: 4 GiB
	• Operating system: 1 GiB
	• LIA: 0.35 GiB
	• SDC: 6 GiB
	• NVMe target: 9 GiB
	• NVMe initiator: 2 GiB
	• SDR: 8586 MiB + 550 MiB * (max remote PDs count)
	NOTE: For certain system configurations, the SDS must be configured to be able to use more than one NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM. By default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0 (usually half the total memory) is less than the memory required by the SDS, you must allow the SDS access to the memory in the other NUMA.
	Non-volatile RAM (SDPM):
	• Node storage capacity less than 41.9 TiB: SDPM size 32 GiB
	• Node storage capacity 41.9 TiB to 91 TiB: SDPM size 64 GiB
	• Node storage capacity greater than 91 TiB and less than 112 TiB: SDPM size 96 GiB
	Or non-volatile RAM (NVDIMM):
	• Node storage capacity less than 40.75 TiB : NVDIMM size 32 GiB
	• Node storage capacity 49 TiB to 83.82 TiB: NVDIMM size 64 GiB
	• Node storage capacity greater than 83.82 TiB and less than 111.76 TiB: NVDIMM size 96 GiB
	The NVDIMM and NVDIMM battery replacement feature are available in PowerFlex Manager for PowerFlex appliance and PowerFlex rack storage-only nodes.
	NOTE: PowerFlex does not support using SWAP on storage-only and storage VM (SVM) PowerFlex nodes. The default setting should be the following:
	• Using SWAP with PowerFlex has the following impact:
	  ○ Slows the performance of the system
	  ○ Impacts I/O on the system disks which are used for SWAP; which might cause other functions using the system disk such as to have slow I/Os:
	  ▪ MDM repository-related operations
	  ▪ PowerFlex logs written to disk. See how to How to Permanently Disable Swap Partition in Linux. In addition, the settings for vm.overcommit must be set to the following settings on nodes that have MDM and SDS processes on them: `in/etc/sysctl.conf` file
	`vm.overcommit_memory=2`
	`vm.overcommit_ratio=100`

Table 15. Physical server requirements (continued)

<table>
<tr><th>Component</th><th colspan="6">Requirement</th></tr>
<tr><td></td><td colspan="6">For more information about the implication of this setting, see the Linux overcommit strategies.</td></tr>
<tr><td rowspan="3">Operating system or boot disk</td><td>Disk type</td><td>RAID</td><td>Endurance</td><td>Minimum capacity</td><td>Sustained performance IOPS (random 4 KB)</td><td>Sustained performance bandwidth (sequential)</td></tr>
<tr><td>BOSS card</td><td>2*M.2 devices in RAID 1</td><td>0.5 DWPD over 5 years or 200 PB life span writes</td><td>960 GB</td><td>50/20K read/ write</td><td>400/80 [MB/sec] read/write</td></tr>
<tr><td colspan="6">NOTE:
Due to a limitation, do not install any of the following on the same operating system disk as the MDM: PowerFlex Manager and PowerFlex Installer.
The customer-provided infrastructure for PowerFlex management (KVM/Linux) disk recommendation is for reference purposes only. Enhance performance by selecting high-performance read/write speed storage solutions such as SSDs or NVMe drives.</td></tr>
<tr><td>Disk space</td><td colspan="6">Disk space formula per node: 1 GB * number of nodes in the system + 1 GB (for log collection)
Required path locations: <code>/opt</code>
1 GB for log collection under <code>/tmp</code> (reused every time a log bundle is created)
For SDC running on Linux or Citrix Hypervisor (XenServer)—1 GiB
NOTE:
Only current customers who have installed XenServer or Citrix Hypervisor are supported, new customers are not advised to install XenServer or Citrix Hypervisor.
The minimum operating system size for a three-node cluster is 10 GiB.
For VMware ESXi—640 GiB is the minimum for a boot disk; the PowerFlex SVM resides on the boot datastore.</td></tr>
<tr><td>Connectivity</td><td colspan="6">One of the following:
10/25/40/50/100 GbE network
Dual-port network interface cards (recommended)
Ensure the following:
There is network connectivity between all components.
Network bandwidth and latency between all nodes is acceptable, according to application demands.
Ethernet switch supports the bandwidth between network nodes.
MTU settings are consistent across all servers and switches.
The TCP ports are not used by any other application, and are open in the local firewall of the server.
NOTE: You can change the default ports.</td></tr>
</table>

NOTE:

- If buying a PowerEdge based solution with a PowerFlex software license, select hardware that is identical to the engineered solution (PowerFlex appliance, PowerFlex rack, PowerFlex custom node).
- PowerEdge configurations that differ from the engineered solution are at risk of encountering issues. Aligning to the engineered solution is critically important when selecting drives on an SDS device.
- Avoid agnostic drives. Select vendor-specific drives to ensure the nodes ship with PowerFlex validated drives.
- To ensure PowerFlex software compatibility, it is recommended to configure a solution using a PowerFlex base (PowerFlex appliance, PowerFlex rack, PowerFlex custom node).

Supported operating systems

PowerFlex storage nodes support SUSE Linux Enterprise Server (SLES) 15.6 and VMware ESXi 8.0 for NVMe over TCP client. In addition, storage data client (SDC) compute-only nodes support: VMware ESXi operating systems 8.0 and 7.0. The latest patches of VMware ESXi and vCenter for compute-only nodes are tested periodically.

VMware has discontinued support for specific CPUs. For information, see CPU Support Deprecation and Discontinuation in vSphere Releases.

For the most up-to-date list of supported operating systems, see the Dell PowerFlex Simple Support Matrix for this release.

Technology preview features are not fully supported and may not be functionally complete. These features are not suitable for deployment in PowerFlex systems. However, these features are provided as a courtesy and the primary goal is for the features to gain wider exposure with the goal of full support in the future. Customers are encouraged to provide feedback and functionality suggestions for a technology preview feature before it becomes fully supported.

Required Linux packages

The following table lists the packages that must be installed on each PowerFlex node running Red Hat Enterprise Linux or SUSE Linux Enterprise Server (SLES) operating systems.

Table 16. Required packages

PowerFlex component/node	Requirement
All components for both Red Hat Enterprise Linux and SLES	• numactl • libaio1 • wget • apr • libapr1 • python2-rpm or python3-rpm • yum-utils
MDM components	• bash-completion (for SCLI completion) • Latest version of Python 3.x (for newer operating systems such as CentOS/RHEL 8.x) • binutils • java-11-openjdk-headless • java-17-openjdk-headless
SDS nodes	• smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils
Diagnostic tools	• sar • sysstat
PowerFlex-supported storage controllers	• Dell PERCCLI • HP HPSSACLI
SDS components that contain NVDIMMs	• ndctl • jq • daxio • libpmem
Secure authentication mode	Ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system: • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm`
LDAP	Ensure that OpenLDAP 2.4 is installed on all servers.

Table 16. Required packages (continued)

PowerFlex component/node	Requirement
LIA	pciutils
NVMe initiator	Linux nvme_tcp and nvme_fabrics modules must be loaded in the operating system. Use `lsmod | grep nvme` to validate that they are available.

Converting CentOS to SUSE Linux Enterprise Server operating system

PowerFlex Manager automates the process of converting nodes from an embedded operating system based on CentOS to an embedded operating system based on SLES. This feature is available on PowerFlex appliance and PowerFlex rack systems.

In PowerFlex Manager 4.6.x and above, you can convert hyperconverged and storage-only nodes from CentOS to SLES. This feature only applies to PowerFlex rack and PowerFlex appliance deployments on storage-only deployments or SVM nodes that are managed by PowerFlex Manager.

For PowerFlex nodes in lifecycle mode, contact Dell Technologies Support for details about whether a manual procedure is available for your environment.

Microsoft Windows support

Microsoft Windows is supported only on the PowerFlex storage data client (SDC) and LIA components.

- To install the SDC, ensure that Microsoft Security Update KB3033929 is installed.

To use secure authentication mode, ensure that the following are installed on all servers in the system:

- OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported)
- Visual C++ redistributable 2010 package, 64-bit

The Microsoft Windows Storage Spaces feature (available from Windows 8.1 and later) is not supported in Microsoft failover clusters running on PowerFlex volumes.

Storage virtual memory requirements

The following is a list of PowerFlex storage virtual memory requirements.

Storage virtual memory requirements for Dell PowerEdge servers (R660/R760), storage-only nodes include non-volatile random-access memory (SDPM), and storage-only nodes include non-volatile random-access memory (NVDIMM) Dell PowerEdge servers (R650/R750).

Table 17. Storage virtual memory requirements

Component	Memory allocation rules
Memory	PowerFlex RAM (GiB) • SDS FG: 10 GiB + 550 MiB per 1 TiB device size + 100 MiB per device • SDS MG: 5 GiB + 335 MiB per 1 TiB device size • MD Read Cache: Total_Drive_Capacity_in_TiB * 8% • MDM: 6.5 GiB • CloudLink: 4 GiB • Operating System: 1 GiB • LIA: 0.35 GiB • SDC: 6 GiB • SDR: 8586 MiB + 550 MiB * (max remote PDs count)

Table 17. Storage virtual memory requirements (continued)

Component	Memory allocation rules

	NOTE: SDR storage virtual memory support is only available as part of a PowerFlex appliance or the PowerFlex rack managed by PowerFlex Manager.
	• SDR memory requirement for hyperconverged systems (storage virtual machines): 12 GB
	NOTE: For certain system configurations, the SDS must be configured to be able to use more than one non-uniform memory access (NUMA) domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0 (usually half the total memory) is less than the memory required by the SDS, you must allow the SDS access to the memory in the other NUMA.
	Non-volatile RAM (SDPM):
	• Node storage capacity less than 41.9 TB: SDPM size 32 GiB
	• Node storage capacity 41.9 TiB to 91 TiB: SDPM size 64 GiB
	• Node storage capacity greater than 91 TiB and less than 112 TiB: SDPM size 96 GiB
	or non-volatile RAM (NVDIMM):
	• Node storage capacity less than 49 TiB: NVDIMM size 32 GiB
	• Node storage capacity 49 TiB to 96 TiB: NVDIMM size 64 GiB
	• Node storage capacity greater than 96 TiB: NVDIMM size 96 GiB
	NOTE: PowerFlex does not support using swap on storage-only and storage virtual machine nodes, which is the default setting.
	Using swap with PowerFlex has the following impact:
	1. Slows the performance of the system.
	2. Impacts I/O on the system disks which are used for swap. This might cause other functions that use the system disk to have slow I/Os such as:
	a. MDM repository-related operations
	b. PowerFlex logs written to disk
	Disable swap and set `vm.overcommit` to the following on nodes that have MDM and SDS processes on them in the `in/etc/sysctl.conf` file:
	`vm.overcommit_memory=2`
	`vm.overcommit_ratio=100`

Base storage virtual memory	350 MiB
Tiebreaker MDM	50 MiB

Software-defined persistent memory

Software-defined persistent memory (SDPM) is supported as non-volatile memory on Dell PowerEdge servers (R660/R760), storage-only and hyperconverged infrastructure.

The features of SDPM include:

- Battery backup provides enough power to store the data in memory.
- A vault optimized storage solution to write or store the data from acceleration memory to a persistent location.
- Compression and high-speed memory performance in shared system memory.

SDPM is supported for both storage-only and hyperconverged deployments on PowerFlex R660 nodes and PowerFlex R760 nodes.

Supports discovery and deployment of PowerFlex storage-only, and PowerFlex hyperconverged configurations on PowerFlex R660 nodes and PowerFlex R760 nodes. The SDPM nodes can be discovered using the RedFish API that is integrated with iDRAC. iDRAC allows you to configure the BIOS and allocate SDPM.

Upgrade CloudLink

PowerFlex supports CloudLink version 8.2, a device encryption software.

Upgrade from CloudLink 7.x to CloudLink 8.2.x on existing deployments and install CloudLink 8.2 on new deployments of PowerFlex. For more information about CloudLink documentation and guidelines on how to upgrade from CloudLink 7.x to 8.x, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.

Browser support

PowerFlex supports Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

Other requirements

PowerFlex requires that you use a minimum of three SDS servers and three storage devices (a minimum of one disk per SDS), with a combined free capacity of at least 720 GB. These minimum values are true per system and per storage pool.

5

Architecture

This section presents the main components of the PowerFlex architecture.

PowerFlex software components

The PowerFlex block storage solution consists of the following components:

- PowerFlex metadata manager (MDM): Configures and monitors PowerFlex. The MDMs contain all the metadata that is required for system operation.
- Storage data server (SDS): Manages the capacity of a single server and acts as a backend for data access. The SDS is installed on all servers contributing storage devices to PowerFlex. These devices are accessed through the SDS.
- Storage data client (SDC): A lightweight device driver that exposes PowerFlex volumes as block devices to the application that resides on the same server on which the SDC is installed.
- Storage data target (SDT): The NVMe target. It manages the translation of I/O from the internal PowerFlex protocol to the NVMe over TCP standard.
- NVMe initiator: NVMe driver that connects to the NVMe over TCP storage system to consume storage, using an NVMe multipathing driver for high availability and load balancing.
- Storage data replication (SDR): A component that acts as a proxy between the SDS and SDC that must be installed for asynchronous replication.

Depending on the preferred configuration, the software components are installed on the server and create a virtual SAN layer that is exposed to the software applications that reside on the client servers.

PowerFlex metadata manager cluster

The metadata manager (MDM) is the monitoring and configuration agent for PowerFlex. It is the authority that controls and tracks data storage ownership, mapping, data protection, managing rebuilds and all system-related functions. No I/Os run through the MDM.

The MDM is involved in:

- Capacity allocation
- Spare allocation
- Data migration
- Rebuild and rebalance

A metadata manager is any server with the PowerFlex MDM package that is installed on it. A cluster of three or five MDMs is installed on different servers to support high-availability. The same MDM package is installed on all members of the cluster. But one MDM is assigned the primary MDM role, and the others act as secondary or tiebreaker MDMs, depending on the role assigned to them during deployment.

The cluster is created automatically, and roles are assigned using the automated deployment tools during PowerFlex deployment. Roles cannot be changed later without reinstalling the MDM.

PowerFlex also supports a single-node role in which the MDM is installed only on one server. Dell Technologies does not recommend using single-MDM mode in PowerFlex systems, except in temporary situations., because the MDMs contain all the data that is required for system operation. Single mode has no protection and exposes the system to a single point of failure.

Each MDM role functions as follows:

- The primary MDM controls the storage nodes, SDCs, and storage data target (also known as the NVMe target). The primary MDM also contains and updates the MDM repository, the database that stores the storage node configuration and determines how data is distributed between the nodes in the system. This repository is constantly replicated to the secondary MDMs, so that they can take over with no delay. Every MDM cluster has one primary MDM.
- The secondary MDMs are the MDMs in the cluster that are ready to take over the primary MDM role, if necessary. A three-node cluster contains one secondary MDM, which allows for a single point of failure. A five-node cluster contains

two secondary MDMs, allowing for two points of failure. This increased resiliency is a major benefit to enabling the five-node cluster.

- The sole role of the tiebreaker is to help determine which MDM is the primary MDM in the event of MDM failover. A three-node MDM cluster contains a single tiebreaker, whereas a five-node cluster contains two tiebreakers. This ensures that there are always an odd number of MDMs in a cluster, which guarantees that there is always a majority in electing the primary MDM.

The secondary MDMs are also designated "replica" MDMs—meaning that each one contains a copy of the primary MDM repository. The replica allows a nearly instantaneous switchover to a secondary MDM in the event of primary MDM failure.

Each MDM has a unique MDM ID, and can be assigned a unique name.

The following figure illustrates a five-node MDM cluster:

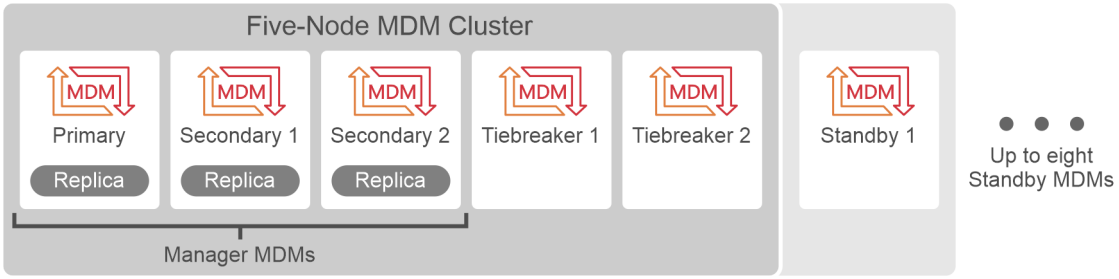


Figure 1. Five-node MDM cluster

Each PowerFlex deployment can also contain one or more standby MDMs, which are not a part of the cluster. In the event of MDM node failure, the standby MDM can be promoted to a cluster role. You can prepare standby manager and tiebreaker nodes, for a total of thirteen cluster and standby MDMs. A standby manager can be called on to assume the position of the primary or a secondary MDM in the cluster. A standby tiebreaker can be promoted to a tiebreaker member of the cluster.

The MDM cluster is limited to 16 IP addresses, which include all cluster members (primary, secondary, standby MDM, and standby tiebreaker).

The following table describes the available MDM cluster modes:

Table 18. MDM cluster modes

Cluster mode	Members	Description
Three-node (default)	• Primary MDM • Secondary MDM • Tiebreaker	Three-node cluster has two copies of the repository, thus can withstand one MDM cluster failure.
Five-node	• Primary MDM • Two secondary MDMs • Two tiebreakers	Five-node cluster has three copies of the repository, thus can withstand two MDM cluster failures.
Single-node	Primary MDM	Single-node cluster has only one copy of the repository, thus it cannot withstand failure. **NOTE:** Dell Technologies does not recommend using single mode in PowerFlex systems.

Virtual IP addresses for the MDM cluster

You can define up to four virtual IP addresses for the MDM cluster. SDCs are then mapped to the virtual IP addresses, instead of to static MDM IP addresses. MDMs are sometimes switched during normal operation of the cluster, and the virtual IP address will always be mapped to the active MDM.

The use of virtual IP addresses simplifies maintenance procedures on the MDM cluster, because system components communicate via the virtual IP addresses. Therefore, SDCs do not need to be reconfigured when a server hosting an MDM is replaced.

In new installations in Linux environments, the virtual IP address of the MDM cluster can be added and mapped using the CSV installation file. In VMware environments, virtual IP addresses are mandatory, and configuration is performed using the PowerFlex VMware Installation Wizard, in the Configure SVM stage. The REST API can also be used to add virtual IP addresses to the cluster. In all cases, a virtual IP NIC placeholder must be mapped to each virtual IP address. Ensure that NICs are available for this purpose.

Existing systems may be extended to include additional MDMs in a cluster. The new MDMs should be mapped to the existing virtual IP addresses.

If virtual IP addresses need to be modified, you must use the PowerFlex CLI (SCLI) or the REST API, instead of the CSV installation file or the vSphere plugin, exercising extreme caution. All SDCs then require reconfiguration to reflect the changes made to the MDM cluster. Otherwise, the SDCs cannot communicate with the MDM cluster, and volumes are not accessible.

A maximum of four MDM virtual IP networks are supported.

System stability

The MDM proxy ability augments the network stability capabilities.

In the PowerFlex architecture, each storage data client (SDC) is required to be fully connected to the MDM for control and management of actions. If an SDC is unable to reach the MDM, the SDC moves to disconnected state, which at some point could lead to complete I/O failure.

If a network failure occurs, PowerFlex allows clusters that do not use the MDM virtual IP address capability to use the secondary MDMs as a proxy for all control traffic to provide greater redundancy.

If a network failure occurs and the MDM proxy is in use, the management API raises an appropriate alert.

Installing the MDM in multiple-rack deployments

The following list includes best practices for adding MDMs to a multiple-rack system:

- For a large system with multiple racks, the MDM cluster members should be distributed in separate racks for resiliency.
- The MDM network requirements that are described in the *Networking* section must also apply to the network connections between the racks.
- You can ensure higher resiliency when the rack is shut down for planned or unplanned reasons, especially if fault sets are used. For more information, see *Fault sets*.
- To ensure that users do not shut down the MDMs by mistake, mark them physically with stickers saying **Do not shutdown: MDM installed here**, or add a meaningful prefix or suffix to the hostname of the server where the MDM is installed. For example, *Rack22U32-MDM*. You can also confirm that a different node in the rack is used per server as shown in the following table:

Table 19. Nodes in rack

Node level	Rack number	MDM role
Top node	1	MDM
Bottom node	2	MDM
Middle node	3	TB

Related information

Networking
Fault sets

Storage data servers

The storage data server (SDS) is installed on Linux-based servers contributing storage devices to the storage cluster. Each SDS is a PowerFlex node.

An SDS manages the capacity of a single server and acts as a backend for data access. Working together, the storage data servers in the cluster abstract local storage, maintain storage pools, and present volumes to the SDCs and NVMe hosts.

PowerFlex systems can scale from a few SDS nodes to hundreds in a cluster, linearly scaling I/O performance, throughput, and storage capabilities.

Storage components

The PowerFlex architecture includes the following storage elements: protection domains, fault sets, and storage pools. Together, these elements link the physical PowerFlex layer to the virtual storage layer.

Protection domains

A protection domain is a logical entity that consists of a group of SDSs that back up each other. Each SDS belongs to one protection domain, so each protection domain is a unique set of SDSs.

The maximum recommended number of nodes in a protection domain is 128. This enables the following:

- Optimal performance
- Reduction of theoretical *mean time between failure* issues
- Ability to sustain multiple failures in different protection domains

You can add protection domains during installation. In addition, you can modify protection domains post-installation with all the management clients (except for OpenStack) .

Fault sets

A fault set is a logical entity that contains a group of SDSs within a protection domain.

In a large cluster that spans across several racks, a fault set can be used as a means to increase the overall cluster resiliency. If one of the racks incurs power loss the use of a fault set prevents a denial of availability. By grouping SDSs in different fault sets, a verification process occurs so that two copies of data do not reside in the same fault set which increases the cluster resiliency. When the data is written to the SDS device, two copies on two SDS in different fault sets are verified.

When defining fault sets, the term *fault units* refers to either a fault set, or an SDS not associated with a fault set.

There must be enough capacity within at least three fault units to enable mirroring.

If fault sets are defined, you can use any combination of fault units, for example:

- SDS1, SDS2, SDS3
- FS1, SDS1, SDS2
- FS1, FS2, SDS1
- FS1, FS2, FS3

In the following figure, there are three protection domains. The middle protection domain (fully depicted) consists of seven SDSs, each with two storage devices.

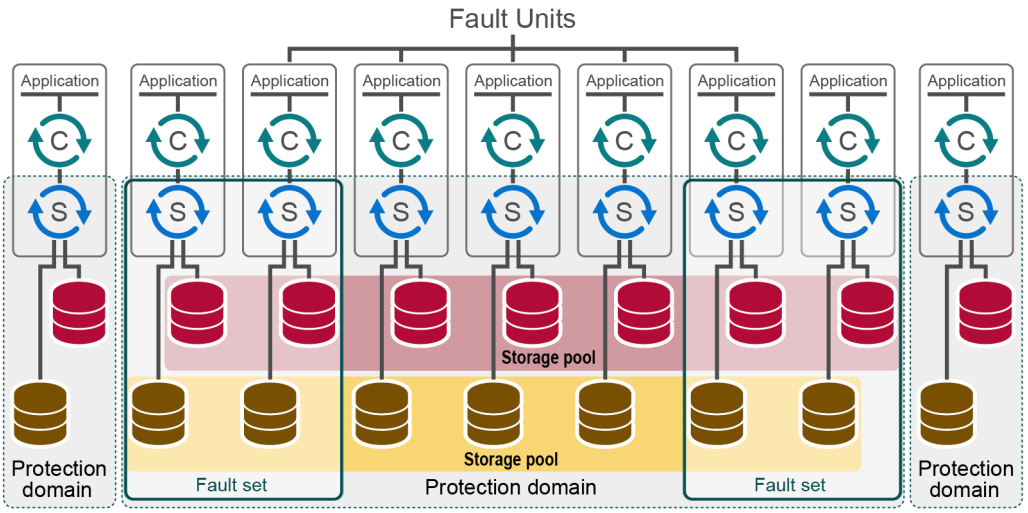


Figure 2. Protection domains, storage pools, and fault sets

Complete the following steps to use fault sets:

1. Ensure that a protection domain exists, or add a new one.
2. Ensure that a storage pool and fault sets (a minimum of three fault units) exist, or add new ones.
3. Add the SDS, designating the protection domain and fault set, and simultaneously, adding the SDS devices into a storage pool.

   The automated deployment and installation tools follow this order automatically.

   You can only create and configure fault sets before adding SDSs to the system, and configuring them incorrectly may prevent the creation of volumes. An SDS can be added to a fault set only during the creation of the SDS.

You can also add fault sets when adding SDS nodes after initial installation.

Related information

Installing the MDM in multiple-rack deployments
PowerFlex implementation in a VMware ESXi-based system

Storage pools

A storage pool is a set of physical storage devices in a protection domain. A volume is distributed over all devices residing in the same storage pool. You can define a magnetic storage pool (for HDDs) or a high-performance storage pool (for SSDs). Storage pools support medium (1 MB) or fine (4 KB) granularity data layouts and allow enabling or disabling zero padding (the process of zeroing out all the bits of an SDS device).

Storage pools allow the management of different storage tiers in PowerFlex. Each storage device belongs to one storage pool.

When a volume is configured over the virtualization layer, it is distributed over all devices residing in the same storage pool. Each volume block has two copies on two different SDSs. This virtualization layer allows the system to maintain data availability following a single-point failure. If a fault set is configured, then each copy will also be on different fault sets. Two network failures render all the domains in a state where I/O errors are possible on any storage pool.

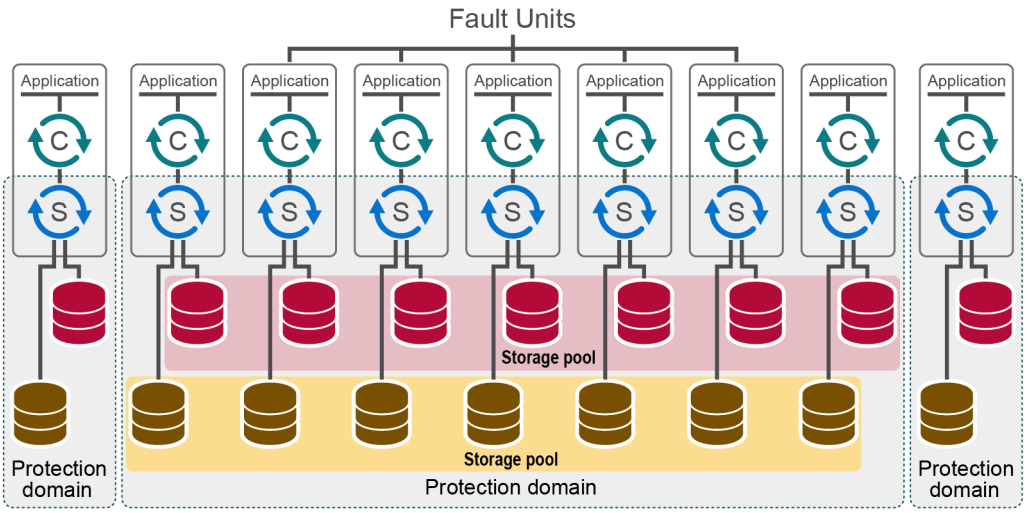


Figure 3. Protection domains and storage pools

You must assign a media type setting to each storage pool. Supported types are: HDD, SSD, and transitional (allows for migration flows).

If all SDSs in a protection domain have two physical drives associated with them, such as one hard drive, and the other SSD, you should define two storage pools:

- Magnetic storage pool

  Consists of all HDDs in the protection domain

- High-performance storage pool

  Consists of all SSDs used for storage purposes in the protection domain

  NOTE:
  - Mixing different types of SSDs is not recommended. Creating a separate storage pool for each type is the recommended best practice, for example: SAS SSD, SATA SSD, NVMe SAS SSD.
  - PowerFlex might not perform optimally if there are large differences between the sizes of the fault units in the same storage pool. For example, if one device has a much larger capacity than the rest of the devices, performance may be affected. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI `modify_sds_device_capacity` command.

Storage pools support the following data layouts for HDD or SSD media:

- Medium granularity:
  - Space allocation occurs at 1 MB units
  - Supports HDD and SSD media
  - Includes persistent checksum for data integrity
- Fine granularity:
  - Requires SSD media and SDPM or NVDIMM for acceleration
  - Space allocation occurs at 4 KB units
  - Includes persistent checksum for data integrity
  - Supports data compression which reduces the size of data that is stored on the disk
  - Supports thin-provisioned, zero-padded volumes

NOTE: Fine granularity and medium granularity storage pools can both exist in a single SDS. You can also migrate volumes across the two layouts.

Each storage pool can work in one of the following modes:

- Zero-padding enabled

Ensures that every read from an area previously not written to returns zeros. Some applications might depend on this behavior. Furthermore, zero padding ensures that reading from a volume will not return information that was previously deleted from the volume.

This behavior incurs some performance overhead on the first write to every area of the volume since the area must be filled with zeros first.

Fine granularity is always zero padded.

- Zero-padding disabled (default only for medium granularity)

  A read from an area previously not written to will return unknown content. This content might change on subsequent reads.

Some applications assume that when reading from areas not written to before, the storage will return zeros or consistent data. If you plan to use such applications, then zero padding must be enabled.

You can add storage pools during installation. In addition, you can modify storage pools post installation with most of the management clients.

NOTE: The zero padding policy cannot be changed after the addition of the first device to a specific storage pool.

Larger PowerFlex storage-only nodes

The maximum SDS is 160 TB (node) on medium granularity storage pools.

- SSD maximum is 15.36 TB (SSD drives)
- Allows higher density systems with larger capacity storage nodes (SDS)
- Initially limited to medium granularity data layout storage pools only (no expansion)

Acceleration pools

Acceleration pools require non-volatile random-access memory for fine granularity storage pools.

Non-volatile random-access memory acceleration pool features and requirements:

- Must reside on SDPM or NVDIMM devices.
- The SDPM or NVDIMM must be mapped to a DAX device.
- Non-volatile random-access memory acceleration pools cannot be disabled.
- Migration to a different NVRAM pool is not supported.

Limitation: SDPM or NVDIMM failure on SDS node causes failure of all SSDs in the fine granularity storage pool for that node.

An acceleration pool can serve multiple storage pools, but a single storage pool cannot use multiple acceleration pools.

An acceleration device can serve multiple SDS devices, but SDS devices cannot span across multiple acceleration devices.

Software-defined persistent memory

Software-defined persistent memory (SDPM) is supported as non-volatile memory on Dell PowerEdge servers (R660/R760), storage-only and hyperconverged infrastructure.

The features of SDPM include:

- Battery backup provides enough power to store the data in memory.
- A vault optimized storage solution to write or store the data from acceleration memory to a persistent location.
- Compression and high-speed memory performance in shared system memory.

SDPM is supported for both storage-only and hyperconverged deployments on PowerFlex R660 nodes and PowerFlex R760 nodes.

Supports discovery and deployment of PowerFlex storage-only, and PowerFlex hyperconverged configurations on PowerFlex R660 nodes and PowerFlex R760 nodes. The SDPM nodes can be discovered using the RedFish API that is integrated with iDRAC. iDRAC allows you to configure the BIOS and allocate SDPM.

Data layout

PowerFlex offers a space efficient storage layout with fine granularity.

With fine granularity, since space allocation occurs at 4 KB, the data that is stored on the disk is significantly reduced. Space allocation for medium granularity occurs at 1 MB.

For fine granularity, you must have SDPM or NVDIMM configured as a DAX device. After SDPM or NVDIMM is all set up in your environment, you must configure an SDPM or NVDIMM acceleration pool and then you can select it in the storage pool.

Data-compression is not supported for medium granularity. It supports thin provisioned, zero padded volumes. Zero padding is enabled by default for fine granularity and cannot be disabled. Zero padding is disabled for medium granularity.

Fine granularity supports persistent checksum for data integrity.

Volumes

You must add and configure volumes before enabling applications to access them.

The adding and mapping volume process is necessary, as part of the getting started process, before applications can access the volumes. In addition, you may create additional volumes and map them as part of the maintenance of the virtualization layer.

You can configure the caching option when creating the volumes, or you can change the read RAM cache feature later. If you want to enable the caching feature, ensure that the feature is also enabled in the backend of the system, for the corresponding storage pool and SDSs.

Define volume names according to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query, or via the property sheet for the object in PowerFlex Manager. It is highly recommended to give each volume a meaningful name associated with its operational role.

Migrating volumes

You can migrate volumes to a different storage pool or protection domain within your PowerFlex system.

You can migrate volumes from the following storage layers:

- From one storage pool to another within the same protection domain
- From one storage pool to another across different protection domains
- From the volume tree and all of its snapshots when they are migrated together
- From thick to thin
- From fine or medium to medium or fine granularity

Requirements to migrate volumes:

- Medium granularity pools should have zero padding enabled.

Snapshots

PowerFlex enables you to create snapshots, instantaneous copies of volumes. A snapshot can be manipulated as any volume exposed by the storage system. Snapshots that are taken simultaneously form a consistency group and can be manipulated as a group.

The PowerFlex storage system enables you to take snapshots of existing volumes, up to 128 per volume. The snapshots are thin provisioned and are extremely quick.

Once a snapshot is generated, it becomes a new, unmapped volume in the system. You can manipulate it in the same manner as any other volume exposed by the PowerFlex storage system.

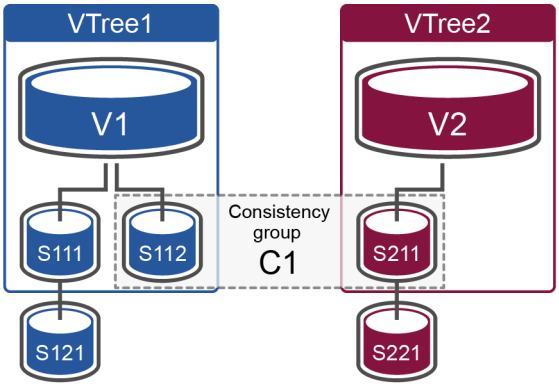


Figure 4. Snapshot operations

The structure related to all the snapshots resulting from one volume is referred to as a vTree (short for volume tree). When taking a snapshot in the system, you can specify more than one volume. All snapshots that are taken together form a consistency group. So if there is a contextual relationship between the data that is contained by all the snapshot members, then that set is meaningful. The consistency group allows manipulation of the entire set.

If you remove an entire consistency group, all snapshots that were taken together are removed. In RED, S211 is a snapshot of V2. Since S112 and S211 were taken together, they compose a consistency group that is designated as C1.

NOTE:

- PowerFlex does not implement any protection mechanisms to preserve the consistency group. For example, you can remove a snapshot that is a member of a consistency group.
- The system supports removing a parent snapshot into a child snapshot.
- It is recommended not to trim volumes that contain snapshots as the logical capacity may not shrink.

Related information

SAN virtualization layer
Snapshot policies

Volume trees (vTrees)

A vTree consists of the root volume and all of its descendant volumes and snapshots.

A vTree is the structure comprised of a volume and the snapshots resulting from that volume. It is a tree spanning from the source volume at its root, whose descendants are either snapshots of the volume itself or snapshots of a snapshot. In the vTree diagram, S_{111} and S_{112} are snapshots of V_1. S_{121} is a snapshot of snapshot S_{111}. Together, V_1 and S_{1xy} are the vTree of V_1.

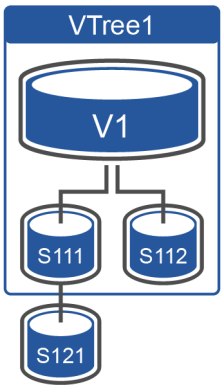


Figure 5. vTree diagram

Snapshot policies

Snapshot policies enable you to define policies where you can configure the number of snapshots to take at a given time for one or more volumes.

Snapshots are taken according to the rule defined. You can define the time interval in-between two rounds of snapshots and the number of snapshots to retain, in a multilevel structure. For example, take snapshots every x minutes/hours/days/weeks. There are one to six levels, with the first level having the most frequent snapshots.

Example:

Rule: Take snapshots every 60 minutes

Retention levels:

- 24 snapshots
- 7 snapshots
- 4 snapshots

After defining the parameters, select the source volume to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per vTree may be used as a source volume of a policy (any policy). When you remove the source volume from the policy, you must choose how to handle snapshots. Snapshots created by the policy are referred to as auto snapshots. In PowerFlex Manager, this is indicated if a snapshot policy is displayed for the snapshot. For more information, see Snapshots.

- If the source volume has no auto snapshots, you cannot unassign the source volume from the snapshot policy. You can remove auto snapshots from **Snapshots** in PowerFlex Manager.
- If the source volume has auto snapshots but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are just detached from the snapshot policy, but not deleted. You must manually remove auto snapshots.

NOTE: Associating replicated volumes that belong to a replication consistency group with a snapshot policy isn't supported.

Related information

Add an RCG snapshot policy
Snapshots

Hosts

PowerFlex supports both storage data clients (SDCs) and NVMe hosts to map and present volumes to the operating system. SDCs and NVMe hosts can co-exist in the same cluster; the only restriction is that they cannot use the same volume simultaneously.

The SDC is a PowerFlex-proprietary lightweight device driver that provides frontend volume access to applications, and maps and presents volumes to the operating system or hypervisor. SDCs maintain peer-to-peer connections to every SDS managing a pool of storage. They translate SCSI to PowerFlex protocol.

NVMe hosts consume PowerFlex block storage using NVMe over Fabrics (NVMe-oF) connectivity to map and present volumes to the operating system or hypervisor. From a user perspective, NVMe hosts play the same role in the system as SDCs. However, they co-exist on the SDS and do not require any additional hardware.

The following figures show the difference in connectivity between the SDC and the NVMe host. The SDC sits on the application host or compute node and connects directly to the storage layer. The NVMe host, on the other hand, connects to an NVMe target called the storage data target, which runs on the storage side. The storage data target directs the volumes onward to the storage components.

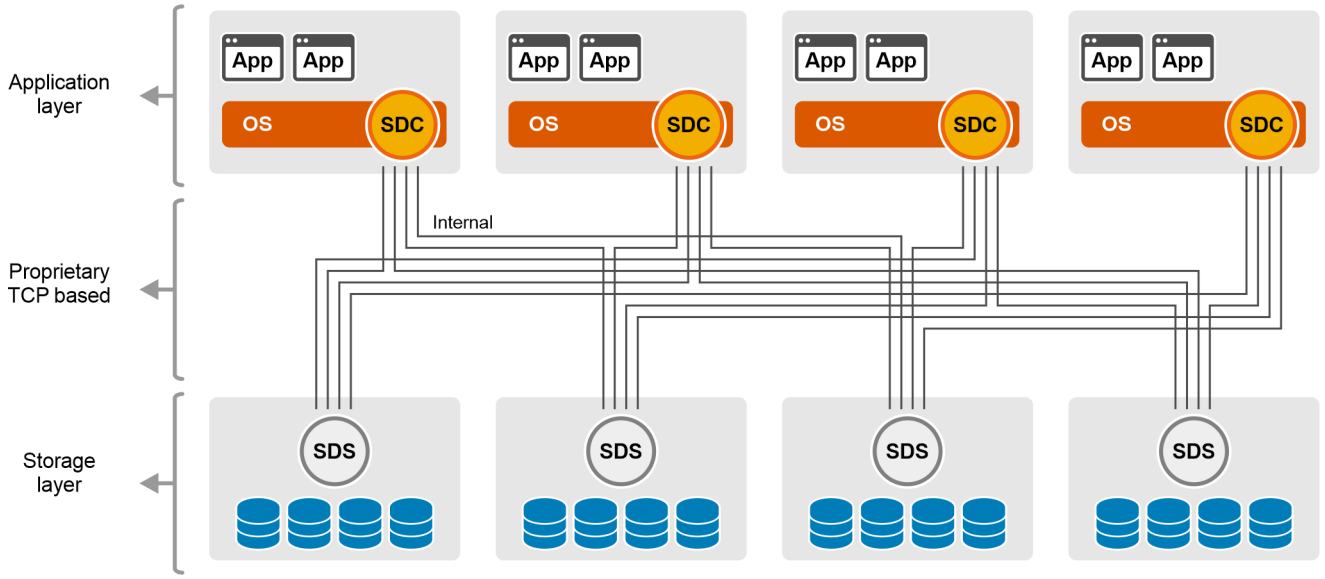


Figure 6. SDC host connectivity

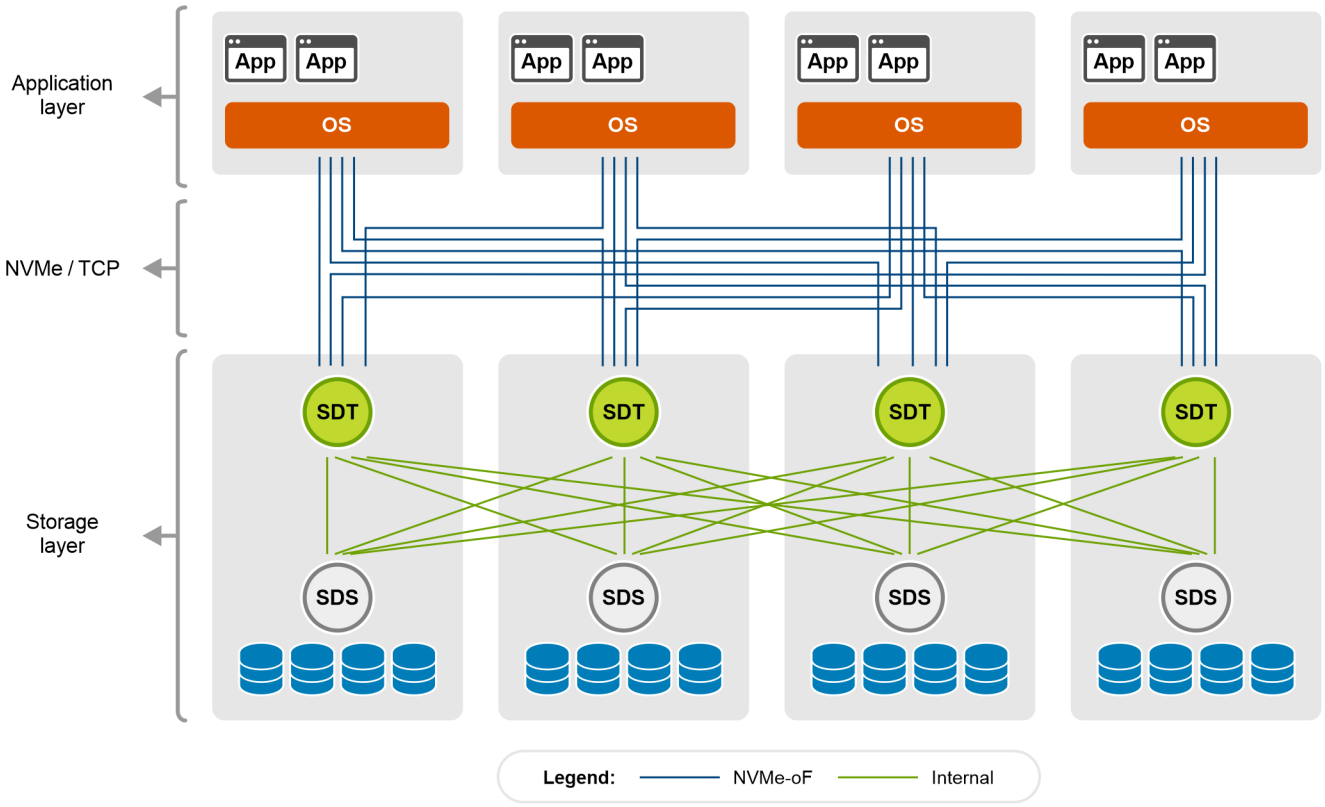


Figure 7. NVMe over TCP host connectivity

Storage data clients

The storage data client is a driver installed on hosts that translates the PowerFlex control and data path protocols to the SCSI layer. The SDC handles all the PowerFlex multipathing aspects and connects to all the PowerFlex storage components.

The SDC also integrates with relevant operating system logic to provide a native block storage device in the operating system. Volumes (also know as LUNs in other storage systems) are mapped to the SDC to be consumed as block devices.

The unified SDC versioning package simplifies storage data client deployment and lifecycle management.

SDCs are supported on Linux, Windows, AIX, and other operating systems, and can be installed on any non-PowerFlex host meeting PowerFlex specifications for SDCs. PowerFlex appliance and PowerFlex rack also support installing SDCs in hyperconverged configurations.

SCSI support

The SCSI identity consists of the following:

- The VMware ESXi SDC registers a new SCSI adapter with Fibre Channel transport.
- The Fibre Channel node worldwide name is the low 64 bits of the SDC GUID.
- The Fibre Channel port worldwide name is the high 64 bits of the SDC GUID.
- The Fibre Channel port and node worldwide names are also exposed for the virtual Fibre Channel targets.

Every mapped volume contains the following:

- A SCSI worldwide name which can be queried at a vital product data (VPD) page `0x83` EUI-64 designator using the SCSI Inquiry command. The EUI-64 designator is 128 bits, and it is composed of the PowerFlex system MDM ID and Volume ID.
- A SCSI unit serial number can be queried though a VPD page `0x83` EUI-64 designator using the SCSI Inquiry command. For example:

```
MOS_SNPRINTF((char *)&pUnitSerial->serialNum[0],
serialNumBufSize, "%s-%.16"MOS_X64_BIT__STRING"-%.16"MOS_X64_BIT__STRING,
MOS_VERSION__LEGACY_COMPANY_NAME_CAP, mapVol_GetMdmID(pVol), mapVol_GetVolID(pVol));
```

  The serial number follows the pattern `EMC - MDM_ID-Vol_ID`.

Every mapped volume is assigned a Target LUN address when it is mapped. Because the storage I/O (SIO) volume ID is 64 bits, the address is calculated as follows:

- The SDC exposes up to 8 K LUNs in total. Begin with the volume index (low 32 bits of the volume ID) and use the modulo operation with 8 K. The result is the SCSI volume index.
- The low eight bits of the SCSI volume index become the LUN ID, and the remaining top five bits becomes the target ID. If there is a collision, which is when a volume exists with a TargetLun pair, the LUN ID stays a low eight bits and the target ID increments by one until the collision is resolved. If a collision exists for all 32 target IDs, the LUN ID increments by one and starts over until the collision is resolved.
- SDC reports a single target worldwide name that is based on the SDC GUID. This target ID is unrelated to the target that is part of the volume LUN ID.

Supported SCSI commands

The following tables describe the SCSI commands that PowerFlex supports.

Table 20. Mandatory SCSI commands

Command type	Operation code	Comments
FORMAT UNIT	04h	N/A
INQUIRY	12h	INQUIRY commands support a subset of the standard pages.
READ	READ (10): 28h READ (12): A8h READ (16): 88h READ (32): 7Fh/0009h	PowerFlex supports various sizes of READ commands: READ (10), READ (12), READ (16), and READ (32).
READ CAPACITY (10)	25h	N/A
READ CAPACITY (16)	9Eh/10h	N/A
REPORT LUNS	A0h	N/A
TEST UNIT READY (TUR)	00h	N/A

Table 21. Optional SCSI commands

Command type	Operation code	Comments
COMPARE AND WRITE	89h	Atomic test and set (ATS)
MODE SENSE - only page `0xa` / subpage `0x0`	1Ah	MODE SENSE commands support a subset of the standard pages.
Reservations related		
PERSISTENT RESERVE IN	5Eh	SCSI-3 reservation support
PERSISTENT RESERVE OUT	5Fh	SCSI-3 reservation support'
RESERVE	0x16	SCSI-2 reservation support. PowerFlex supports RESERVE 1 and RESERVE 6.
RELEASE	0x17	SCSI-2 reservation support. PowerFlex supports RELEASE 6 and RELEASE 10.
UNMAP	42h	Thin provisioning or block delete
WRITE	WRITE (10): 2Ah WRITE (12): AAh WRITE (16): 8Ah WRITE (32): 7Fh	PowerFlex supports different sizes of WRITE commands: WRITE(10), (WRITE 12), WRITE (16), and WRITE (32)
WRITE SAME	WRITE SAME (10): 41h WRITE SAME (16): 93h WRITE SAME (32): 7Fh	Zero block or write zeroes. PowerFlex supports different sizes of WRITE SAME commands: WRITE SAME (10), WRITE SAME (16), WRITE SAME (32)

The following command is not supported:

- EXTENDED COPY

Supported Vital Product Data pages

PowerFlex supports the following pages for SCSI Vital Product Data (VPD):

- `0x00` - Standard inquiry
- `0x80` - Unit serial number
- `0x83` - Device identification
- `0x80` - Block limits
  - This page includes size limitations used by the optional commands: Unmap, Write Same, and Compare and Write.
- `0x83` - Logical block provisioning

NVMe over TCP overview

NVMe over TCP is based on NVMe over Fabrics (NVMe-oF), which is an industry-standard protocol for storage transport. NVMe-oF is supported by multiple operating systems and does not require any host agent on the storage node.

TCP provides a standard, common, interoperable solution that delivers excellent performance. The TCP transport allows the use of existing networks and hardware.

PowerFlex hosts can use standard NVMe over TCP operating system drivers to connect to PowerFlex. They do not require proprietary host agents. These drivers are available in mainstream Linux distributions and in VMware ESXi.

NVMe over TCP is supported in a disaggregated, two-layer PowerFlex deployment architecture. The NVMe host is deployed on a compute node and communicates directly with the NVMe target, which is known as the storage data target (SDT), on the storage node. NVMe hosts are deployed through PowerFlex Manager.

PowerFlex supports NVMe over TCP on the following operating systems:

- VMware ESXi 7.0U 3f (Dell Technologies recommends using the latest release) and VMware ESXi 8.0 (Dell Technologies recommends using the latest release)
- SUSE Linux Enterprise Server, SLES 15 SP6, with the x86_64 with 6.x.

  All other Linux operating systems are in technology preview state and can be run in testing scenarios.

  NOTE: Technology preview features are not fully supported and may not be functionally complete. These features are not suitable for deployment in PowerFlex systems. However, these features are provided to the customer as a courtesy and the primary goal is for the features to gain wider exposure with the goal of full support in the future. Customers are encouraged to provide feedback and functionality suggestions for a technology preview feature before it becomes fully supported.

For the up to date list, see the NVME/TCP Host/Storage Interoperability Simple Support Matrix.

PowerFlex supports NVMe reservations (SCSI-3 equivalent) with NVMe over TCP.

NVMe hosts

PowerFlex supports connecting and consuming storage from NVMe hosts. An NVMe host is a host running an operating system that supports NVMe over TCP.

After an NVMe host is created in PowerFlex, you can map volumes to the host so that the NVMe host can consume storage. The NVMe host connects to an NVMe target called the storage data target, which run on the storage side. The storage data target directs the volumes onward to the storage components.

The following host parameters are configurable:

- Number of connections. For hosts that are expected to generate high I/O load, you can increase the default value.
- Number of paths per volume. Up to eight paths can be configured per volume in the host attribute. This is mostly an operating system limit per volume and per total paths.

PowerFlex supports specifying networks that might fail together. NVMe over TCP connections are allocated to the NVMe over TCP hosts through multiple network sets, which offers resiliency to the failure of a network set.

NVMe over TCP supports fault sets. Connections allocated to the NVMe over TCP host are through targets on multiple fault sets that provide path resiliency to a fault set failure.

Storage data target (NVMe target)

The storage data target (SDT) is the channel for communications with NVMe hosts. It serves as the NVMe target and translates NVMe over Fabrics protocol into internal PowerFlex protocol.

The storage data target is deployed with the SDS on each storage server and provides access to the volumes inside that protection domain.

In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports assigned to that host, either by the automatic load balancing or by manual connectivity configuration.

Persistent discovery

PowerFlex supports persistent discovery for NVMe over TCP targets. When NVMe hosts are initially discovered through the persistent discovery controller, they remain connected to the discovery service. This enables automatic updates of the hosts regarding storage side changes.

When a change in the discovery information is provided to the NVMe host, the discovery controller returns an asynchronous event notification and the host requests the updated discovery log page.

Changes in discovery information include the following:

- A new volume is mapped to the host from a new protection domain.
- A new storage data target is added to the system.
- Load balancing has moved the host connection from one storage data target to another storage data target.

Before you use this feature, confirm that your host operating system supports the persistent discovery controller. Ensure that the persistent discovery controller is enabled on the host. Supported operating systems include VMware ESXi 8.0, Red

Hat Enterprise Linux 9, and SLES 15.4 or higher. For more information, see the NVMe over TCP host configuration in the documentation of the respective operating system.

Load balancing

PowerFlex supports NVMe over TCP connectivity and adds NVMe-oF targets to enable hosts to connect to storage. When hosts are allowed to connect and consume storage, the system ensures that it remains resilient, balanced, and available.

PowerFlex NVMe over TCP load balancer

Use the optimal PowerFlex system performance by distributing the NVMe host workloads over the NVMe targets (storage data targets). PowerFlex offers two ways to achieve load distribution: manual and automatic. In both cases, system ports are selected for the host and are returned to the host when it performs NVMe discovery.

The automatic load balancing, which is the default policy uses an algorithm that selects the system ports based on the following:

- Access to the mapped volumes of the host
- Path resiliency
- Load balancing

The host connectivity plan is recalculated under the following conditions:

- Host connects.
- Storage data target is added or removed.
- Network set or system network is modified.
- Storage data target is added or removed to a fault set.
- Storage data target IP address is reassigned to a different network.

NOTE: The load balancer ignores temporary issues such as network failures or if a PowerFlex storage-only node goes down for a few minutes.

The storage system provides multiple ports through which the host can connect, though every port and node has a limited throughput. To achieve the best performance, ensure that the combined I/O load of the hosts is distributed well over the available system ports and nodes. The NVMe over TCP load balancer aims to automate the planning and execution of the host-to-storage connectivity task to achieve path resiliency and workload balance.

The PowerFlex load balancer considers the host networks for resiliency. When a host loses a network, it can still use other paths through available networks to access the storage.

NVMe over Fabrics discovery

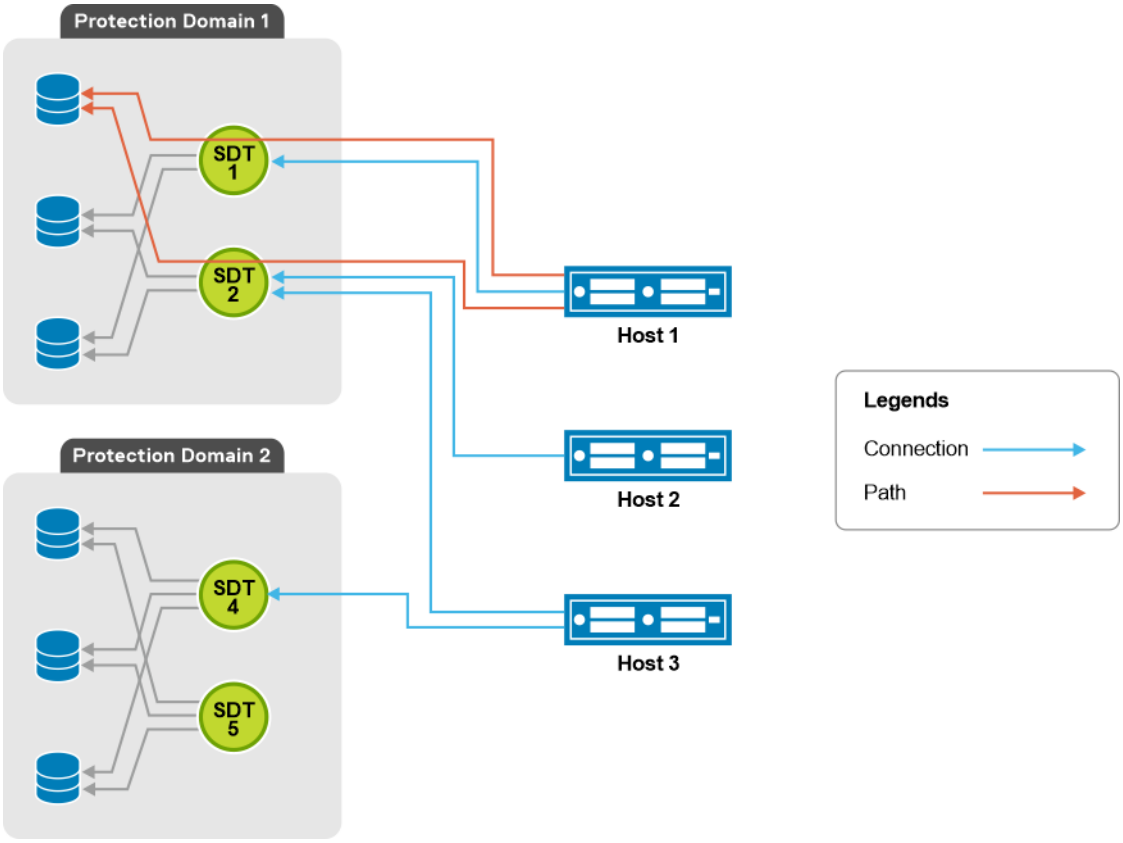
NVMe over Fabrics includes a discovery protocol, which is a standard method for the host to discover and connect to available subsystems.

The host connects to a discovery service and receives discovery information, including all the connection ports to access its volumes. PowerFlex continues to update the host regarding changes using the persistent discovery controller. In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports that are assigned to that host by the load balancer. The host can also be configured manually, If the user opted for the manual connectivity configuration.

Balancing host connections

When a host is mapped to the first volume from a protection domain, it assigns ports that might connect to volumes in the protection domain. Each port represents a connection. The number of ports that are assigned is a configurable host attribute, which is up to 128 per protection domain. The system selects the least occupied ports to ensure fault isolation. The number of paths per volume is a configurable host attribute, which is a maximum of eight paths per volume. This configurable host attribute is mostly an operating system limit per volume and total paths.

NVMe over TCP connections and paths



A connection is the establishment of the NVMe controller and includes the following:

- Link between a host part and a specific storage data target IP address.
- Connection couple (host-port and system-port).
- The load balancer provides 10 ports for each host by default, and up to 128 ports.

A path is a connection that is used to provide access to a volume. This connection makes the path a tuple (host-port, system-port, and volume), which must be a subset of the connections.

By default, the load balancer provides four paths per volume to each host, which is selected from the assigned ports.

Manual load balancing

Users might prefer to manually configure host connectivity. Manual configuration is appropriate for special cases, such as when hosts generate a higher-than-expected traffic load.

To configure manually, the user designates in the host configuration the system ports to which the host must connect.

Migration to NVMe over TCP

PowerFlex provides the option to migrate workloads from SDC to NVMe over TCP on VMware ESXi: the migration can be performed either online or offline.

- Online migration using storage vMotion (VMFS only): The standard way of moving storage is with storage vMotion. Storage vMotion also supports switching protocols by migrating to a new datastore. For more information, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.
- Offline migration (VMFS only): There is an option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without sending all the data over the network. For more information about the offline migration steps, see SCSI to NVMe VMware VMFS Datastore Offline Migration Steps.

NOTE: There are no standard ways to migrate Linux environments, VMware ESXi clusters, and RDMs.

Storage data replicator

Storage data replicators process all I/Os of replication volumes.

The source storage data replicators processes all application I/Os of replicated volumes. At the source, the SDC sends application I/Os to the storage data replicator. The I/Os are sent to the target storage data replicators and stored in their journals. The journals of the target storage data replicators apply the I/Os to the target volumes. A minimum of two storage data replicators are deployed at both the source and target systems to maintain high availability. If one storage data replicator fails, the MDM directs the SDC to send the I/Os to an available storage data replicator.

The source storage data replicator processes all application I/Os of replicated volumes. The source storage data replicator saves changes to the journal, and later sends them to the target storage data replicators. At the target, the storage data replicators receive the changes, and when a consistent image is available, apply the changes to the target volumes.

A minimum of two SDRs are deployed at the source and target protection domain to ensure high availability.

The ratio of SDRs in a source protection domain to the storage data replicators in its target protection domain is limited. The maximum ratio of peer storage data replicators is 1:3, and the maximum ratio of storage data replicators in maintenance is 1:6. The `Add_SDR` and `Remove_SDR` commands fail if the operation results in an unsupported ratio. If the storage data replicators ratio exceeds the allowed value as a result of a failure or planned maintenance, the `Add_RCG` command fails. Existing replication consistency groups and replication operations are not impacted.

Protection and load balancing

PowerFlex maintains user data in a distributed mesh mirrored layout. Each piece of data is stored on two different fault units. The copies are distributed over the storage devices according to an algorithm that ensures uniform load of each fault unit by terms of capacity and expected network load. Rebuild and rebalance processes are fully automated, but are configurable.

Rebuild

PowerFlex initiates a rebuild process in response to failure. Forward rebuild refers to creating a new copy of the data on another server. Backward rebuild refers to re-synchronizing one of the copies.

When a failure occurs, such as on a server, device or network failure, PowerFlex immediately initiates a process of protecting the data. This process is called *rebuild*, and comes in two flavors:

- *Forward rebuild* is the process of creating another copy of the data on a new server. In this process, all the devices in the storage pool work together, in a many-to-many fashion, to create new copies of all the failed storage blocks. This method ensures an extremely fast rebuild.
- *Backward rebuild* is the process of re-synchronization of one of the copies. This is done by passing to the copy only changes made to the data while this copy was inaccessible. This process minimizes the amount of data transferred over the network during recovery.

PowerFlex automatically selects the type of rebuild to perform. This implies that in some cases, more data will be transferred to minimize the time that the user data is not fully protected.

Rebuild throttling

The rebuild throttling policy determines the priority of rebuild I/Os versus application I/Os when accessing SDS devices. The possible rebuild throttling policies are no limit on rebuild I/Os, limit concurrent I/Os per SDS device, favor application I/Os, and dynamic bandwidth throttling.

Rebuild throttling sets the rebuild priority policy for a storage pool. The policy determines the priority between the rebuild I/O and the application I/O when accessing SDS devices. Note that application I/Os are continuously served.

Applying rebuild throttling will increase the time the system is exposed with a single copy of some of the data, but it will reduce the impact on the application. You must attempt to choose the right balance between the two.

The following priority policies may be applied:

- No limit: No limit on rebuilding I/Os. Any rebuild I/O is submitted to the device immediately, without further queuing.

NOTE: Rebuild I/Os are relatively large and hence setting this policy will speed up the rebuild, but will have the maximal effect on the application I/O.

- Limit concurrent I/O: Limit the number of concurrent rebuild I/Os per SDS device (default). The rebuild I/Os are limited to a predefined number of concurrent I/Os. When the limit is reached, the next incoming rebuild I/O waits until the completion of a currently performed rebuild I/O. This will complete the rebuild quickly for best reliability, however, there is a risk of host application impact.
- Favor application I/O: Limit rebuild in both bandwidth and concurrent I/Os. The rebuild I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebuild I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebuild I/Os will be served. Once either threshold is reached, the rebuild I/Os wait until both I/O and bandwidth are below their thresholds. For example, setting the value to **1** guarantees the device has one concurrent rebuild I/O at any given moment, which ensures the application I/Os only wait for 1 rebuild I/O at worst case. This imposes bandwidth on top of the limit concurrent I/Os option, which is a prerequisite to using this policy.
- Dynamic bandwidth throttling: This policy is similar to favor application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. Note that the limits on the rebuild bandwidth and concurrent I/Os are still imposed.
- Default values:
  - The default policy for rebuild is: Limit concurrent I/O
  - Rebuild concurrent I/O Limit: 1 concurrent I/O

NOTE: Rebuild throttling affects system performance and should only be used by advanced users.

Rebalance

When PowerFlex detects that user data is not balanced across devices in the storage pool, it initiates a process to restore the balance in which data copies are moved from the most used devices to the least used.

Rebalance is the process of moving one of the data copies to a different server. It occurs when PowerFlex detects that the user data is not evenly balanced across the fault units in a storage pool. This can occur as a result of several conditions such as: SDS addition or removal, device addition/removal, or following a recovery from a failure. PowerFlex moves copies of the data from the most used devices to the least used ones.

Both rebuild and rebalance compete with the application I/O for the system resources, which include network, CPU and disks. PowerFlex provides a rich set of parameters that can control this resource consumption. While the system is factory-tuned for balancing between speedy rebuild or rebalance and minimization of the effect on the application I/O, the user has fine-grain control over the rebuild and rebalance behavior.

Rebalance throttling

The rebalance throttling policy determines the priority of rebalance I/Os versus application I/Os when accessing SDS devices.

Rebalance throttling sets the rebalance priority policy for a storage pool. The policy determines the priority between the rebalance I/O and the application I/O when accessing SDS devices. Note that application I/Os are continuously served. Rebalance, unlike rebuild, does not impact the reliability of the system, and therefore reducing its impact is not risky.

NOTE: Rebalance throttling affects the performance of the system, and should be used only by advanced users.

The following possible priority policies may be applied:

- No Limit: No limit on rebalance I/Os. Any rebalance I/O is submitted to the device immediately, without further queuing. Rebalance I/Os are relatively large and hence setting this policy will speed up the rebalance, but will have the maximal effect on the application I/O.
- Limit concurrent I/O: Limit the number of concurrent rebalance I/Os per SDS device. The rebalance I/Os are limited to a predefined number of concurrent I/Os. Once the limit is reached, the next incoming rebalance I/O waits until the completion of a currently performed rebalance I/O. For example, setting the value to **1** guarantees that the device will only have one rebalance I/O at any given moment, which will ensure that the application I/Os only wait for 1 rebalance I/O in the worst case.
- Favor application I/O: Limit rebalance in both bandwidth and concurrent I/Os. The rebalance I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebalance I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebalance I/Os will be served. Once either limiter is

reached, the rebalance I/Os wait until such time that the limits are not met again. This imposes a bandwidth limit on top of the limit concurrent I/Os option.

- Dynamic bandwidth throttling: This policy is similar to favor application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. The limits on the rebalance bandwidth and concurrent I/Os are still imposed.
- Default values:
  - The default policy for rebalance: Favor Application I/O
  - Rebalance concurrent I/O Limit: 1 concurrent I/O per SDS device
  - Rebalance bandwidth limit: 10240 KB/s

For information about instant maintenance mode that uses rebuild and rebalance, see *Instant maintenance mode*.

Related information

Instant maintenance mode

Checksum protection

PowerFlex calculates and validates the checksum value for the payload during transit to protect data in-flight. Checksum protection is applied to all I/Os.

This feature addresses errors that change the payload during the transit through PowerFlex. PowerFlex protects data in-flight by calculating and validating the checksum value for the payload at both ends.

NOTE:

- The feature is off by default for medium granularity.
- The checksum feature can have a significant impact on large block I/O latency. Contact Dell Technologies Support for more information.

- During write operations, the checksum is calculated when the SDC receives the write request from the application. This checksum is validated just before each SDS writes the data on the storage device.
- During read operations, the checksum is calculated when the data is read from the SDS device, and is validated by the SDC before the data returns to the application. If the validating end detects a discrepancy, it initiates a retry. The checksum is done in the granularity of a sector (0.5 KB).
- Pools with fine granularity with or without compression, have a persistent checksum by default. This cannot be changed.

Each I/O goes through compression, and the checksum is calculated before it is written to the disk. There are two types of checksum:

- Fine granularity layout saves checksum data before and after processing to guarantee data integrity (compressed or not).
- There are also system checksums for metadata.

This feature applies to all I/Os: application, rebuild, rebalance, and migrate. The checksum is also kept in RMcache (read memory cache), protecting every block that is maintained in SDS memory against memory corruption. The checksum feature can be enabled at the protection domain level, and defined at the storage pool level. The feature is T10/DIF-ready.

The fine granularity data layout has a default checksum whether it is compressed or not.

Fine granularity layout checksum implementation

The fine granularity data layout has a default checksum whether it is compressed or not.

Cache

PowerFlex supports RAM Read Cache (using DRAM server memory), to enhance system performance.

The caching modes provided by the system are the following:

- Mode: RAM Read Cache (RMcache).
- Description: Read-only caching performed by server RAM.

- Considerations: RAM Read cache, the fastest type of caching, uses RAM that is allocated for caching. Its size is limited to the amount of allocated RAM.

  NOTE: The amount that may be allocated is limited, and can never be the maximum available RAM.

- Default setting: Disabled, except when storage-only nodes are deployed.

Supported platforms:

- PowerFlex
- VxFlex Ready Node with Dell PowerEdge R640, Dell PowerEdge R740xd, or Dell PowerEdge R840 servers

Configure direct attached storage

When adding direct attached storage to SDS, it is recommended to configure the raw devices as stand-alone devices. Upon adding a device to an SDS PowerFlex verifies that the device is clear and returns an error if it is otherwise.

PowerFlex works with any free capacity—internal or direct-attached devices, either magnetic hard disk drives or flash-based devices such as solid state drive (SSD) and PCIe cards. Although PowerFlex can work with any device topology, it is recommended to configure the raw devices as stand-alone devices.

Device data is erased when devices are added to SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. An error is returned, per device, if it is found not to be clear. You can override this check by using the force device takeover option.

The following devices are considered to be not "clear," and thus cannot be added to SDS:

- Linux - A complete device with either a file system or partition, or a partitioned device with a file system.
  - Multipath devices cannot be added as SDS devices.
  - Devices in an LVM group cannot be added to an SDS.
- VMware ESXi - Same as above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices will not be prevented from being added.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

NOTE:

If the server has a RAID controller, PowerFlex prefers to use the caching abilities of the controller for better performance, but is better used when all devices are configured as stand-alone (that is setting each of the devices to RAID 0 separately). Dell Technologies recommends enabling RAID controller caching for hard disk drive devices. Caching is not needed for SSD devices.

NOTE:

For hard disk drives: It is recommended to use RAID controller caching when available as follows:

- READ/WRITE: If the cache is battery-backed
- READ ONLY: If the cache is NOT battery-backed

For SSDs (such as Flash): The setting should be write-through.

6

Replication

PowerFlex enables native asynchronous replication for PowerFlex storage-only and PowerFlex hyperconverged configurations.

Replication can be used to quickly and easily recover from a physical or logical disaster, to migrate data, to test data at a remote site, or to offload backup. The PowerFlex implementation is designed to allow a sub-minute recovery point objective reducing the data-loss to minimal in case of disaster recovery. Replication is elastic and can scale online by adding or removing nodes, flexible, and easy to manage. It enables instant test and failover operations.

Replication peer systems

Replication occurs between two PowerFlex systems, connected by WAN, and designated as peer systems.

PowerFlex supports up to four peer systems from a single source.

NOTE: A source volume can be replicated to just a single target.

Journal

Storage data replicators use the journal to hold the replication data.

The source storage data replicators accumulate the data changes in the journal until sending them to the target. The target storage data replicators accumulate received data in the target journal until a completely consistent image is received and can be applied to the target volumes.

SDR journal capacity

To assign enough storage capacity for the replication journal consider the following factors when allocating journal capacity.

- The storage pools from which to allocate the journal capacity. The journal is shared between all the replicated replication consistency groups in the protection domain. Journal capacity should be allocated from storage pools as fast as (or faster than) the storage pool of the fastest replicated application in the protection domain. It should use the same drive technology and about the same drive count and distribution in nodes.
- The minimal requirements needed, for which the default minimum journal size is 400 GB. The minimum journal size can also be calculated as 28 GB per SDR session (where the number of SDR sessions is the number of SDRs + 1) or the capacity needed to sustain an outage, whichever is greater.
- Expected outage time.
  - The minimal outage allowance is one hour, but at least three hours are recommended when computing the minimum size of the replication journal.
  - Journal capacity needed per application is: **`maximal application throughput`** X **`maximum outage interval`**.
- Journal capacity is calculated as a percentage of storage pool capacity, based on the previously calculated needs. Journal capacity should be at least 5% of replicated usable capacity in the protection domain, including volumes used as source and targets.

For example:

- An application generates 1 GB/s of writes.
- The maximal supported outage is 3 hours (3 hours x 3600 seconds = 10800 seconds).
- The journal capacity needed for this application is 1 GB/s x 10800 s = ~10.547 TB.
- Since the journal capacity is expressed as a percentage of the storage pool capacity, divide the 10.547 TB by the size of the storage pool, which is 200 TB: 100 x 10.547TB/200TB = 5.27%. Round this up to 6%.
- Repeat this for each application being replicated.

When a protection domain has several storage pools and several replicated applications, the journal capacity should be calculated as in the example above, and the capacity can be divided among all the storage pools (provided they are fast enough). For higher availability, the journal capacity should be allocated from multiple storage pools.

NOTE: When the storage pool capacity is critical, capacity cannot be allocated for new volumes or for expanding existing volumes. This behavior must be taken into account when planning the capacity available for journal usage. The volume usage must leave enough capacity available in the storage pool to allow provisioning of journal volumes. The plan should account for the storage pool staying below critical capacity even when the journal capacity is almost fully used.

It is important to note that since journal capacity is defined as a percentage of the total storage capacity in the storage pool, increasing the total storage capacity by adding devices will increase the journal capacity. Similarly, if you decrease the total storage capacity by removing devices from the storage pool, the journal capacity will automatically decrease.

Replication consistency group

The replication consistency group is a logical container for volumes whose application data must be replicated consistently to each other.

For example, if you replicate a database and its transaction log, you need the remote copy of each of the database volumes to reflect an image from the same point in time. You can achieve this by placing all the database volumes in one replication consistency group.

Replication is always defined in the scope of a protection domain. All the objects that participate in the replication are contained in the protection domain, including the volumes in a replication consistency group. The journal capacity from the storage pool in a protection domain is shared among all the replication consistency groups in the protection domain. The SDRs in the protection domain manage I/Os directed to replicated volumes within the protection domain.

The following replication attributes are among those that are defined in the replication consistency group:

- Recovery point objective—The maximum data loss (in units of time) that you are willing to lose. When you set the recovery point objective, this is the goal of replication.
- Direction—Local to remote or remote to local. When there is no replication, for example, following failover or switchover, the direction attribute indicates the direction of the last replication.
- Abstract state—State of replication, for which the options are as follows:
  - **OK**—Replication is healthy, and the recovery point objective target is met.
  - **Recovery point objective violation**—Replication is healthy, but the recovery point objective target is not met.
  - **Error** —Replication is not healthy. The **Error** state attribute provides additional information about the cause for the error.
  - **Stopped by user**—Replication is not carried out due to user action, such as pause or terminate.
- Error state—Describes the error that is preventing data replication.
- Pause mode—When the replication consistency group is paused, all application I/Os are stored in the source journal. When replication of the replication consistency group is resumed, the source SDR sends the journal contents to the target SDR to be applied to the target volumes. You may want to pause a replication consistency group to handle a network issue between the peer systems or when fixing a hardware issue.
- Activity state—When the replication consistency group is active, replication is enabled. When the replication consistency group is inactive, replication is stopped, but the replication consistency group configuration is maintained. The replication consistency group can be automatically inactivated by the system (due to insufficient journal capacity), or can be terminated manually by the user. When activating an inactive replication consistency group, the replication consistency group starts replication from the beginning with a full initialization.
- Failover type—The failover operation requested by the user. For more information, see *Accessing target volumes*.
- Failover state—The states are transient, except for None (no failover) or Done (the failover operation has completed).

Related information

Accessing target volumes

Replication pairs

A replication pair is a pair of volumes with one volume at the source system and one at the target system. The data from the source volume is replicated to the target volume.

The replication consistency group can contain several replication pairs. A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the replication pair have the same attributes (including zero padding and granularity), not doing so can impact performance and capacity.

Recovery point objective

The recovery point objective is the maximum data loss, which is measured in units of time that would be lost in a disaster or outage.

PowerFlex uses journal-based replication to achieve very low recovery point objective and ensure minimal data loss. To ensure recovery point objective compliance, PowerFlex replicates at least twice for every recovery point objective period. For example, setting the recovery point objective to one minute means that PowerFlex can immediately return to operation at the target system with loss of only one minute of data. In order to achieve a recovery point objective of one minute, replication takes place at least every 30 seconds. Recovery point objective compliance is calculated according to when the application data arrives at the target journal. Achieving the recovery point objective is the most important consideration in replication management. The user-defined recovery point objective is the goal of replication. Even so, recovery point objective compliance is not guaranteed. The system reports recovery point objective compliance for each replication consistency group.

Data protection enables replication to multiple systems from one source system to up to four target systems.

Replication I/O flow

For replicated volumes, application data is processed by SDRs at the source and passed to the SDRs at the target.

Application I/Os (both reads and writes) intended for replication volumes are sent from an SDC to an SDR. The source SDR packages the data and distills it so that only the most recent writes are included. The source SDR sends the data over the WAN to the target SDR. At the target system, the SDR processes the replicated data and applies it to the volumes.

Replication auto-provision

PowerFlex supports automatic creation and mapping of a volume in the peer system.

For a given volume on the source system, you can:

- Create a volume on the destination system.
- Create a replication pair for the given source volume and the new destination volume.

In addition, you can request to map the destination volume to an SDC.

Replication initial copy

When replication is first activated for a replication consistency group, the target volumes must be synchronized with the source volumes.

For each replication pair, the entire contents of each source volume are copied to the corresponding target volume when replication is active. This occurs when the replication consistency group containing the replication pair is activated, or when the replication pair is added to an already active replication consistency group. When there is more than one replication pair in the replication consistency group, the order in which the volumes are synchronized is determined by the order in which the replication pairs were activated. You can manually override this order, and also pause and resume the initial copy process, using CLI commands.

Initial synchronization runs while applications perform I/O. The system sends writes to synchronized areas of the volume to the journal. The system ignores writes to areas not yet synchronized. The synchronization process copies the updated content later.

The system limits the number of volumes that undergo **Initial Copy** at any point in time. New volume pairs added to existing replication consistency groups are given priority over new replication consistency groups. This means that replication pairs in a newly activated replication consistency group may not start **Initial Copy** immediately.

Activating or terminating a replication consistency group

When a replication consistency group is active, replication is enabled. When a replication consistency group is inactive, replication is stopped and changes are not tracked, but the replication consistency group maintains its configuration.

By default, a new replication consistency group is created as active. You may, however, specify that it be created in the inactive state.

When the replication consistency group is in an inactive state, replication is stopped and the journal capacity is released. The I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All replication consistency group configurations are maintained, including source and target roles, pair configuration, and volume access modes.

The replication consistency group can be terminated automatically by the system when the journal capacity is fully consumed. You can also manually inactivate the replication consistency group using the terminate command. When the replication consistency group is terminated, volume I/Os bypass the SDR and go directly to the SDS. You can terminate the replication consistency group at both sites separately. Alternatively, you can terminate the replication consistency group at one site; once connectivity is restored, the peer will automatically terminate the replication consistency group at the other site.

Failover can be performed when the replication consistency group is inactive, however consider that the data at the target may be old. While in failover mode, mappings at both the source and target can be activated and volumes can be written to, just like when the replication consistency group is in an active state. There is no data synchronization when the replication consistency group is inactive. You can activate the replication consistency group only once it is no longer in failover mode.

NOTE: Activating a terminated replication consistency group requires full initialization. The initial copy process may take a long time.

The active or inactive replication consistency group state and the commands to terminate or activate a replication consistency group are only available if both peer systems are running PowerFlex v3.6.x or later.

Pausing replication

Replication can be paused for a given replication consistency group.

While paused, any application I/O is stored in the source journal. Once replication is resumed, the journal contents are sent to the target to be applied to the target volumes.

Using replicated volumes

While replication is active, the target volumes remain inaccessible to the hosts with only limited read-only access permitted. This is done to maintain data integrity and consistency between the two peer systems. The data at the target can be accessed to test replication, create snapshots, and to fail over.

Accessing target volumes

There are several situations in which the target volumes are made accessible to the hosts.

- Fail over the replication consistency group, used especially for disaster recovery. During failover, the application I/Os are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are brought to the newest consistent image available on the destination and their access mode is changed to read-write.

Recovery from a failed over state to a normal operation state can be achieved by either restoring the replication, or reversing it.

- Test a failover of the replication consistency group. The test failover mode provides the application with access to the target volumes without interrupting data synchronization.
- Switch over the replication consistency group. Switchover is a planned failover. The application I/Os are stopped at the source, and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

Target volumes can be accessed in R/W only after failover or switchover has been performed. Once the replication consistency group is in failover or switchover mode, you can decide how to continue with replication:

- Restore replication—This maintains the replication direction from the original source and overwrites all data at the target.
- Reverse replication—This changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

NOTE: Prior to performing the restore or reverse operation, you should consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Related information

Replication consistency group

Replication direction and mapping

Use the table as a reference for replication direction and default access to volumes according to a subsequent replication consistency group operation and action of the PowerFlex system.

The replication involves two peers, system A and system B. When the replication is set up, system A is set as source and system B is set as target. The following replication direction refers to the initial direction A to B and to changes to that direction.

Table 22. Replication direction and mapping

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
Normal	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)
After failover	Reverse/restore Remove	N/A - data is not replicated	By default access to the volume is allowed through the original target (system B) It is possible to enable access through the original source (system A)
After failover + reverse **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After failover + restore **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed only through the source (system A)

Table 22. Replication direction and mapping (continued)

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
After switchover	Switchover/test failover stop/failover Remove	B to A	Access to the volumes is allowed only through the original target (system B)
After test failover	Switchover/test failover/ failover Remove	A to B	Access to the volumes is allowed through both systems (system A and system B)

Setting up replication

You are required to initially prepare the infrastructure to use replication in PowerFlex.

The following is an overview of the necessary steps.

1. Add peer systems, at both the source and target systems. Add the corresponding certificates to each system.
2. Add journal capacity, ensuring sufficient capacity.
3. Add SDRs, at least two per protection domain that has replicated volumes.

Perform the following steps to configure replication:

1. Create the replication consistency group. Define the recovery point objective.
2. Add volume pairs. Volumes in a volume pair must have the same size. It is not necessary but is recommended that they have the same attributes (medium or fine granularity, zero padded, or non-zero padded).

   If both systems are 4.x or higher then the target volumes may be automatically provisioned when the pair is created.
3. If the replication consistency group was created as active (the default), then initial copy is started automatically. Otherwise, the replication consistency group must be activated before the initial copy is started.

For additional information about replication, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.

Snapshot policies and replication

PowerFlex combines local and remote protection by using snapshot policies and replication, enabling data protection and quick recovery from disasters. It uses asynchronous replication to create a remote copy of a volume from one cluster to another, ensuring business continuity. PowerFlex provides a robust data protection solution, allowing for efficient recovery in the event of data loss or system failure.

The process of assigning replicated replication consistency group volumes to a snapshot policy associating the replication consistency group with the snapshot policy, rather than individual volumes. This approach helps avoid potential configuration or setup issues that may arise when assigning individual volumes. The association can be performed at either the source or target replication sites and is compatible with fine granularity storage pools. This method enables a combination of local snapshot protection and remote replication protection, allowing for more granular recovery options

- Assign replication consistency group volumes to a snapshot policy by associating the group with the policy, rather than individual volumes.
- Association can be performed at the source or target replication sites.
- Supported with fine granularity storage pools only.
- Enables combination of local snapshot protection with remote replication protection at source sites.
- Allows for multiple replication recovery images for more granular disaster recovery at target sites.

Add an RCG snapshot policy

Create snapshot policies with a maximum of six retention rules that each specify a retention period and number of snapshots to retain, and then add source volumes from the replication consistency group (RCG) to the policy.

Steps

1. On the menu bar, click **Protection** > **RCGs**.
2. From the RCGs list, select the desired RCG, and click **More Actions** > **Add Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box, assign the policy a name and the time interval between snapshots.
4. Create retention rules:
   a. In the **Retention Rules** area, add the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.

   Up to six retention rules can be created.
5. Optionally, clear the **Read Only** checkbox to cancel read-only access. This checkbox is selected by default.
6. Optionally, select the **Secured** checkbox to prevent the snapshots created by the policy from being modified or deleted before the end of the expiration date.
7. Click **Create** or **Create and Activate** to add the snapshot policy to PowerFlex.
8. In the **Snapshot Policy was successfully added** dialog box, click **Assign volume to snapshot policy**.
9. In the **Select volumes you wish to assign to the policy** window, select the desired volumes, and then click **Assign Policy**.
10. Verify that the operation completed successfully, and click **Dismiss**.

Related information

Snapshot policies

7

PowerFlex management platform

The PowerFlex management platform provides the management and orchestration in a single tool for all PowerFlex functions, from deployment through administration and life cycle management.

The PowerFlex management platform is a management software stack that is based on the Rancher Kubernetes Engine (RKE). It is designed as microservices, with specialized modules working together. Installed across three nodes, the PowerFlex management platform provides high availability, resiliency, and reliability.

A web user interface, PowerFlex Manager, provides ease of access to the PowerFlex management platform. You can also access PowerFlex management platform functionality through the PowerFlex REST API or the PowerFlex CLI (SCLI).

Several PowerFlex management platform deployment configurations are supported, depending on the PowerFlex consumption model. These configurations include installing the PowerFlex management platform on the same server as a PowerFlex storage node in an integrated PowerFlex management (non-KVM) for PowerFlex software or cloud deployments or integrated PowerFlex management (KVM) for PowerFlex appliance deployment configuration, and the PowerFlex management cluster for PowerFlex appliance and PowerFlex rack.

PowerFlex management

PowerFlex management covers all product functionalities, including block storage, hardware management, network management, reporting, and alerts.

- The management can be accessed through a Web UI or REST API.
- PowerFlex Manager handles user management for all systems, including:
  - PowerFlex local users
  - LDAP/AD users
- PowerFlex Manager supports system security, as follows:
  - PowerFlex management platform exposes an HTTPS interface for Web UIs and REST APIs.
  - Users must have an account that is defined in PowerFlex.
  - Upon login, users receive an access token.
  - PowerFlex Manager uses OpenID Connect (OIDC).
  - PowerFlex management platform communication with PowerFlex block storage systems is secured using mTLS.
- PowerFlex management platform is a cluster of three or more self-managed Kubernetes nodes, providing resiliency and high availability.
  - Optimized PowerFlex management platform run time to 60 minutes.
- The following CLI tools are available:
  - A CLI that provides full control over the block storage system.
  - A CLI with expanded support for management, reporting, and alerting.
- Enhanced security strength by using SSH keys and non-root users through sudo in management aspects.

  For deployment and post-deployment procedures, see the applicable versions of the *PowerFlex and PowerFlex Manager Software Install and Upgrade Guide* and *PowerFlex and PowerFlex Manager Software Administration Guide* for configuration guidelines.

- Webhooks support enables sending alerts to the webhooks servers designed for software platforms such as BigPanda.
- Multi-subnet and Multi-VLAN for PowerFlex appliance and PowerFlex rack allows defining multiple subnets and VLANs in a single network.

  The most common implementation example is with a management or vMotion network that spans multiple racks. Each rack is its own subnet and VLAN.

  NOTE: Supported on all networks except for the following: PowerFlex data, vSAN, and NSX external networks.

- Higher flexibility of fault sets defines a subset of nodes in a rack as part of a fault set during deployment and expansion operations.

For example, a rack with 20 PowerFlex nodes can have four fault sets, one for every five nodes, for easier maintenance support. Another option is using fault sets in a full rack configuration, such as one fault set per rack.

- PowerFlex management cluster self-awareness on PowerFlex appliance and PowerFlex rack allows self-awareness of its underlying controller system.
  - Upgrade from end to end and receive alerts from issues in the PowerFlex management controller infrastructure.
- Modular (single) component upgrade changes the package in the PowerFlex rack and PowerFlex appliance catalogs to newer or older versions.

  This upgrade depends on customer cases that want to deviate from the Release Certification Matrix or Intelligent Catalog.

  NOTE: By default this feature is turned off and can only be enabled through an RPQ process. If applicable, contact your Dell Technologies account representative for details.

PowerFlex Manager

PowerFlex Manager is the unified management user interface for all PowerFlex systems. It allows you to deploy and manage new and existing PowerFlex systems, and also includes lifecycle and resource management capabilities.

PowerFlex Manager provides IT operations management for PowerFlex systems. It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. PowerFlex Manager can also be used to deploy and manage new and existing PowerFlex appliance and PowerFlex rack systems.

PowerFlex Manager includes the following tabs:

- A dashboard that enables you to see a summary of performance and PowerFlex appliance, PowerFlex rack, and PowerFlex software inventory data.
- A block tab that provides a user interface for managing PowerFlex block storage.
- A protection tab that provides a user interface for performing PowerFlex storage protection operations.
- A lifecycle tab that allows you to create and clone templates for PowerFlex appliance, PowerFlex rack, and PowerFlex software and deploy them as resource groups (previously known as services).
- A resources tab that allows you to manage all resource types within the system.
- A monitoring tab that allows you to track events, alerts, and jobs running.
- A settings tab that enables you to configure system settings for PowerFlex Manager.

PowerFlex Manager is part of all PowerFlex consumption models.

PowerFlex Manager is designed as microservices, specialized modules working together and based on the Rancher Kubernetes Engine.

Management capabilities

PowerFlex Manager provides the unified interface and full lifecycle mode services. Role-based and scope-based access control enables common Power APIs to improve ecosystem and customer integration of the PowerFlex storage solution.

PowerFlex provides one tool for all consumption models.

- PowerFlex Manager (UX, API, CLI)
  - Unified dashboard
  - REST API for PowerFlex rack, PowerFlex appliance, and PowerFlex software
  - PowerFlex block
  - SDC and NVMe over TCP hosts
- Unified services:
  - Security
  - Single sign-on (SSO) and role-based access control (RBAC)
  - Deployment and lifecycle management
  - Events, alerts, and reporting
  - Logging and metrics collection
  - SNMP, secure connect gateway, SMTP, Dell AIOps (formerly known as CloudIQ)
- Unified RBAC and user management

- No separate users or LDAP configuration
- Single set of roles
- Metadata manager (data-path) does not maintain RBAC for PowerFlex 4.0 and later.

- Single WebUI
- Single REST endpoint for all APIs: Legacy and modern, ISG-standard Power API

NOTE: For login information, see the PowerFlex API documentation on the Dell Technologies Developer portal.

- Events (log) and alerts (state) including metadata manager (MDM) events
- Resilient management stack

Authenticating with a single sign-on and a common identity provider

PowerFlex Manager supports authentication for single sign-on (SSO) using a common identity provider (IdP).

This support relies on a multicloud solution based on Red Hat OpenShift using PowerFlex. With this feature, PowerFlex introduces an OpenID Connect (OIDC) IdP provider type as an alternative to SAML, enabling a seamless user experience leveraging SSO to PowerFlex.

PowerFlex Manager and VMware vCenter automation and integration

PowerFlex Manager has extensive orchestration capabilities with VMware vCenter and ESXi.

For orchestration capabilities, see the following for the supported feature set and dependencies:

NOTE: You must supply the relevant license to support the relevant feature set.

Table 23. PowerFlex Manager and VMware vCenter automation and integration

Feature	Dependency
Deploy and configure VMware ESXi	iDRAC access to a PowerFlex node
Deploy and configure VMware ESXi with persistent memory	iDRAC access to a PowerFlex node with persistent memory (SDPM/NVDIMM)
Create VMware ESXi cluster	VMware vCenter access
Set up VMware ESXi HA and DRS	VMware vCenter access with vCLS enabled
Create and configure VMware vCenter based vSphere distributed switch	VMware vCenter access
Create datastore and deploy VMs	Access to VMware vCenter and VMware ESXi
Service and lifecycle VMware ESXi compute-only nodes active workloads	Access to VMware vCenter and VMware ESXi
Enable or disable SSH to VMware ESXi upon demand	Access to VMware ESXi

Supporting non-bonded networking configurations

PowerFlex Manager supports non-bonded networking options with resource group templates.

Configure a storage-only template to use a non-bonded networking configuration where data networks are not bonded. Configure a compute-only template to work with a non-bonded storage-only deployment. To enable non-bonded networking, PowerFlex Manager allows the same PowerFlex management network to be on two ports on the same interface.

Set up two different types of non-bonded configurations on a resource group template:

- Four data network configuration
  - The PowerFlex management network is selected on all ports in interface1 and interface2. The management network is set on Bond0 for this configuration. This configuration reserves only one PowerFlex management IP address.
  - Data 1 is on port 1 of the first interface (assigned to Eth 0 with no bond).

    - Data 2 is on port 2 of the first interface (assigned to Eth 1 with no bond).
    - Data 3 is on port 3 of the second interface (assigned to Eth 2 with no bond).
    - Data 4 is on port 4 of the second interface (assigned to Eth 4 with no bond).
- Two data network configuration
    - The PowerFlex management network is selected on port 1 in interface1 and interface2. The management network is set on Bond0 for this configuration. This configuration reserves only one PowerFlex management IP address.
    - Data 1 is on port 1 of the second interface (assigned to Eth 0 with no bond).
    - Data 2 is on port 2 of the second interface (assigned to Eth 1 with no bond).

For a non-bonded storage-only template, you must select **Trunk Port** for **Switch Port Configuration** setting, not **Port Channel (LACP enabled)**. For the **Teaming and Bonding Configuration** setting, you typically select **Mode 1** for most deployments.

PowerFlex Manager provides eight different sample templates that support non-bonded network configurations. Four of the templates are for storage-only and four are for compute-only deployments.

- Storage-25gb-4Data-Trunked (replaces existing Storage-Non-Bonding template)
- Storage-100gb-4Data-Trunked
- Storage-25gb-2Host-Data-Trunked
- Storage-100gb-2Host-Data-Trunked
- Compute-Linux-25gb-2Data-Trunked (replaces existing Compute-Linux template)
- Compute-Linux-25gb-4Data-Trunked
- Compute-Linux-100gb-4Data-Trunked
- Compute-ESXi-25gb-2Data-Trunked

High availability of the PowerFlex Manager user interface

PowerFlex Manager is able to recover from most failures in a short amount of time, enabling continuous management and operation of the PowerFlex system.

When a node is not available, a banner is displayed at the top of the screen indicating that the system is undergoing maintenance and you must wait up to 10 minutes for recovery. Once the node returns to normal status, the banner disappears. Whenever the user interface is not able to retrieve data from the backend, PowerFlex Manager displays a message indicating that no data is available due to maintenance. The block, alerts, and events services run replicas of the data to allow the relevant parts of the user interface to remain accessible when a node or pod fails.

Serviceability

PowerFlex Manager provides automated and remote support through integration with Dell Technologies connectivity services, which ensures better alignment with Dell Technologies services initiatives to enhance the user experience.

Registering your device with Dell Technologies connectivity services allows Dell Technologies connectivity services to:

- Collect system state information and telemetry.
- Automate issue detection and support case creation.
- Resolve common issues remotely.
- Provide critical information to Dell Technologies remote support to resolve complex issues.

Deployment options for each management offering

PowerFlex management platform is an integrated part of the PowerFlex system. PowerFlex management platform can be set up in multiple deployment options, regardless of the consumption model.

Depending on which consumption model you choose, you have different management options within the solution stack. PowerFlex Manager manages some aspects of the solution while other elements are customer-managed.

Table 24. Deployment option per offering

Supported offering	Integrated PowerFlex management (KVM)	PowerFlex management cluster (KVM)	Integrated PowerFlex management (non-KVM)	Customer-provided infrastructure for PowerFlex management (KVM/Linux)	Single ESXi PowerFlex management node	PowerFlex management cluster (ESXi)
PowerFlex software	No	No	Yes	Yes	No	No
PowerFlex custom node or VxFlex Ready Node	No	No	Yes	Yes	Yes	No
PowerFlex appliance	Yes	Yes	No	Yes Not available by default. Contact Dell Technologies Support for more information.	Yes	Yes
PowerFlex rack	No	Yes	No	Yes Not available by default. Contact Dell Technologies Support for more information.	No	Yes

NOTE: QCOW2 images are available on the Dell Technologies support site and can be downloaded by the customer for a customer-provided infrastructure for PowerFlex management (KVM/Linux) environment.

Dedicated management cluster model

The PowerFlex management platform can be deployed on dedicated management nodes or clusters running outside of PowerFlex storage-only nodes, PowerFlex compute-only nodes, or PowerFlex hyperconverged nodes.

When you deploy a dedicated management cluster, the PowerFlex management platform manages the complete solution stack.

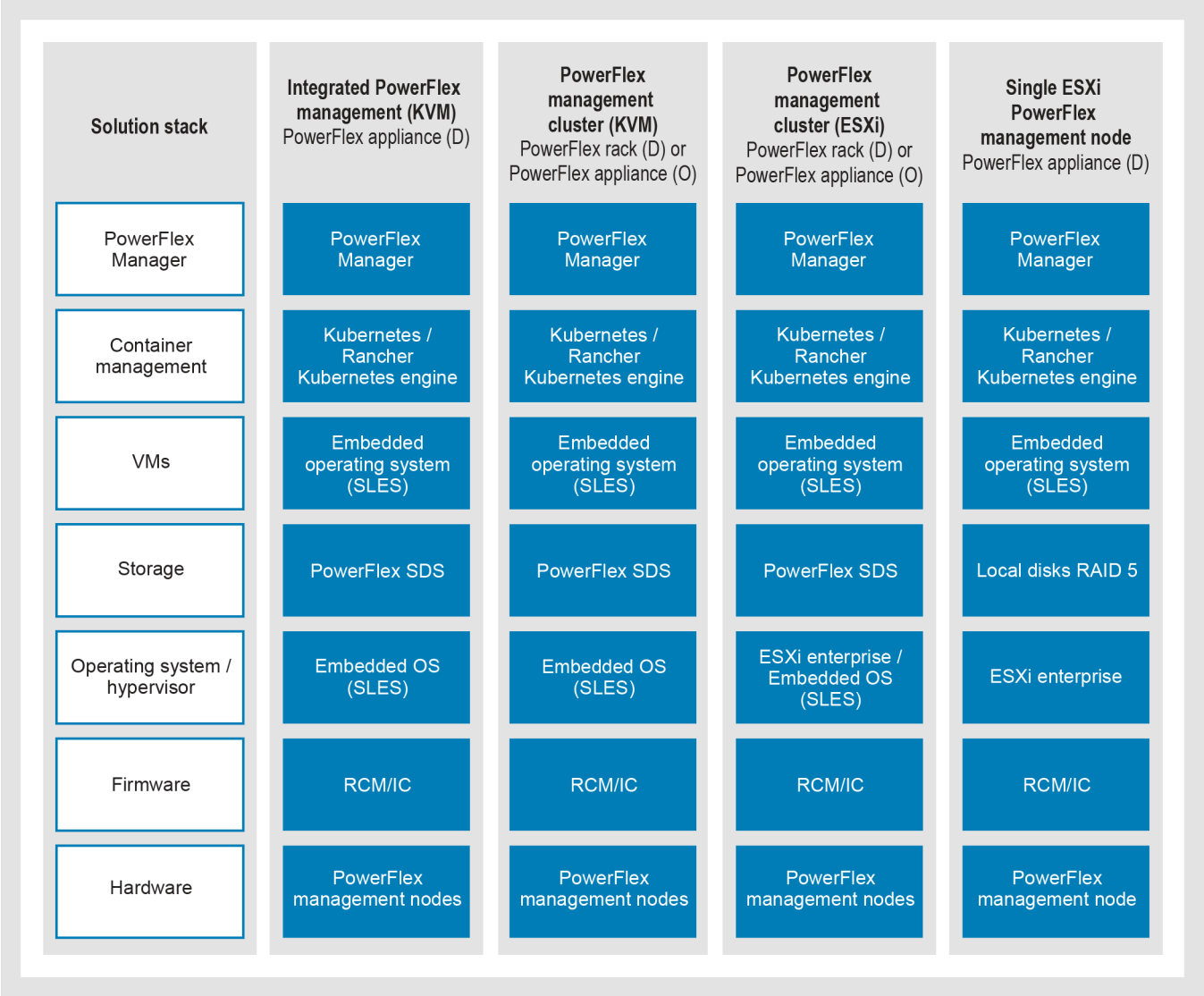


Figure 8. Dedicated management cluster model

Integrated PowerFlex management (non-KVM)

The PowerFlex management platform supports an integrated PowerFlex management (non-KVM) deployment model in which the PowerFlex management platform is deployed on the same machines as PowerFlex storage-only nodes for PowerFlex software and PowerFlex for Public Cloud.

In the integrated PowerFlex management (non-KVM) deployment model, the PowerFlex management platform is deployed on the boot device of three PowerFlex storage nodes so that the PowerFlex management platform and PowerFlex storage co-reside on the same node. For more information, see the *PowerFlex and PowerFlex Manager Software Install Guide*.

In this management option, the PowerFlex management platform manages only the containers and PowerFlex, as illustrated in the following diagram:

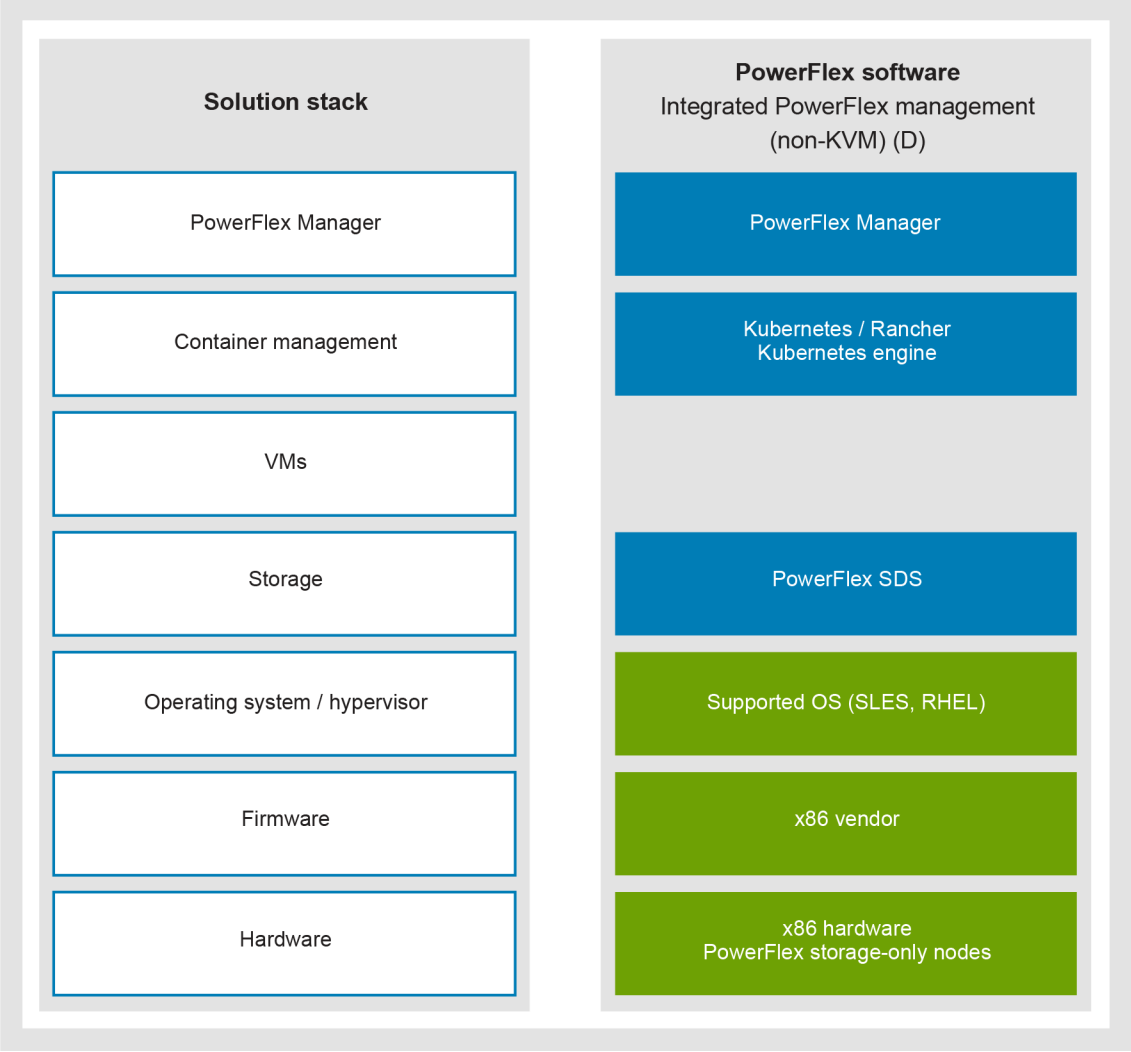


Figure 9. Integrated PowerFlex management (non-KVM) software management model

Integrated PowerFlex management (KVM)

The PowerFlex management platform supports an integrated PowerFlex management (KVM) deployment model in which the PowerFlex management platform is deployed on the same machines as PowerFlex storage-only nodes using a KVM solution for PowerFlex appliance only.

In the integrated PowerFlex management (KVM) deployment model, the PowerFlex management platform is deployed on the boot device of three PowerFlex storage nodes so that the PowerFlex management platform and PowerFlex storage co-reside on the same node. For more information, see the *PowerFlex and PowerFlex Manager Software Install and Upgrade Guide*.

In this management option, the PowerFlex management platform manages only the containers and PowerFlex, as illustrated in the following diagram:

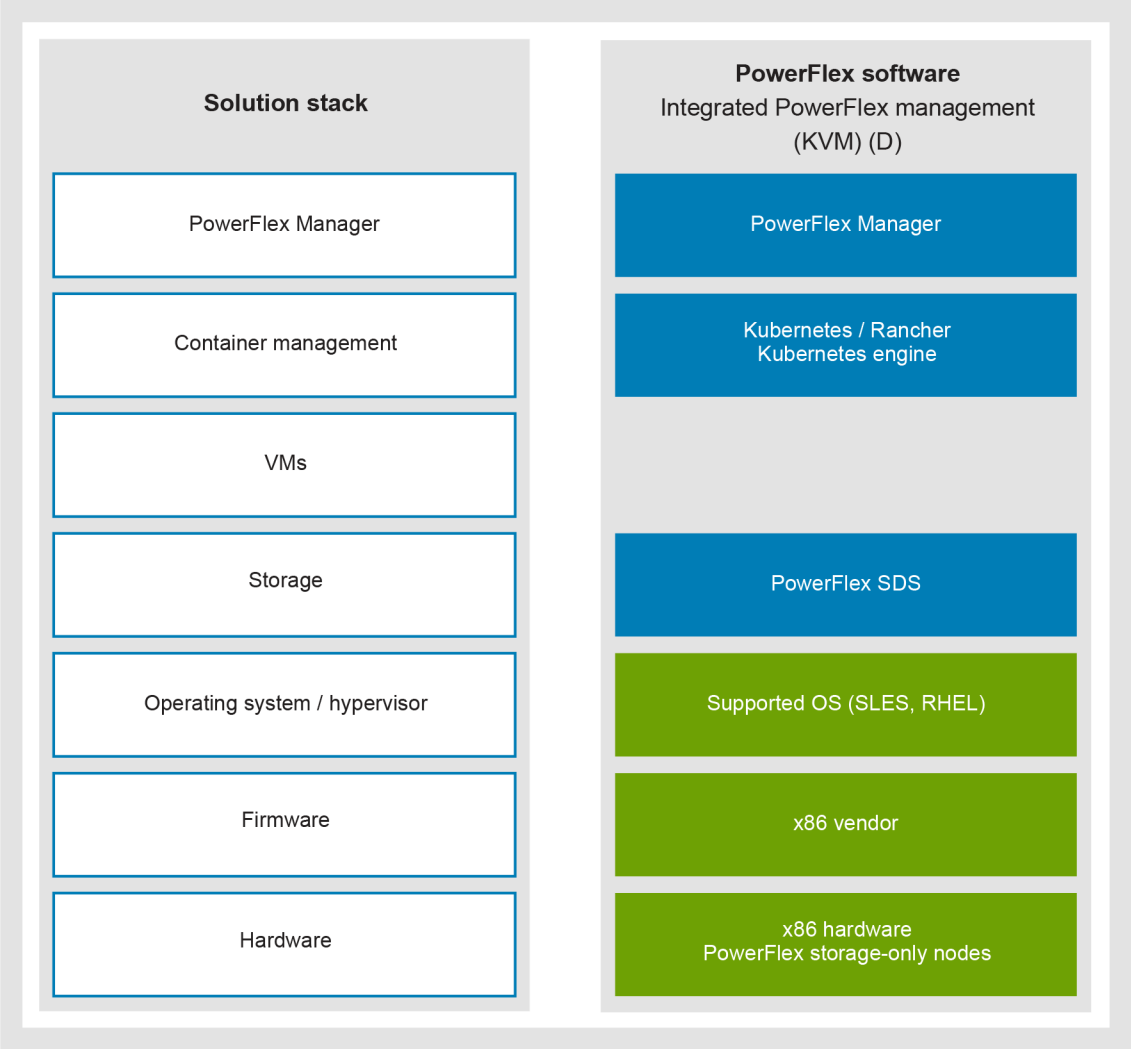


Figure 10. Integrated PowerFlex management (KVM) software management model

Customer-provided infrastructure for PowerFlex management

The PowerFlex management platform can be deployed on a supported operating system of your choice. In this management option, the PowerFlex management platform manages only the containers and PowerFlex.

NOTE: For non-co-integrated PowerFlex management, bring your own KVM models, customers can download QCOW2 images from the Dell Technologies support site.

The following image explains the solution stack, depending on the operating system:

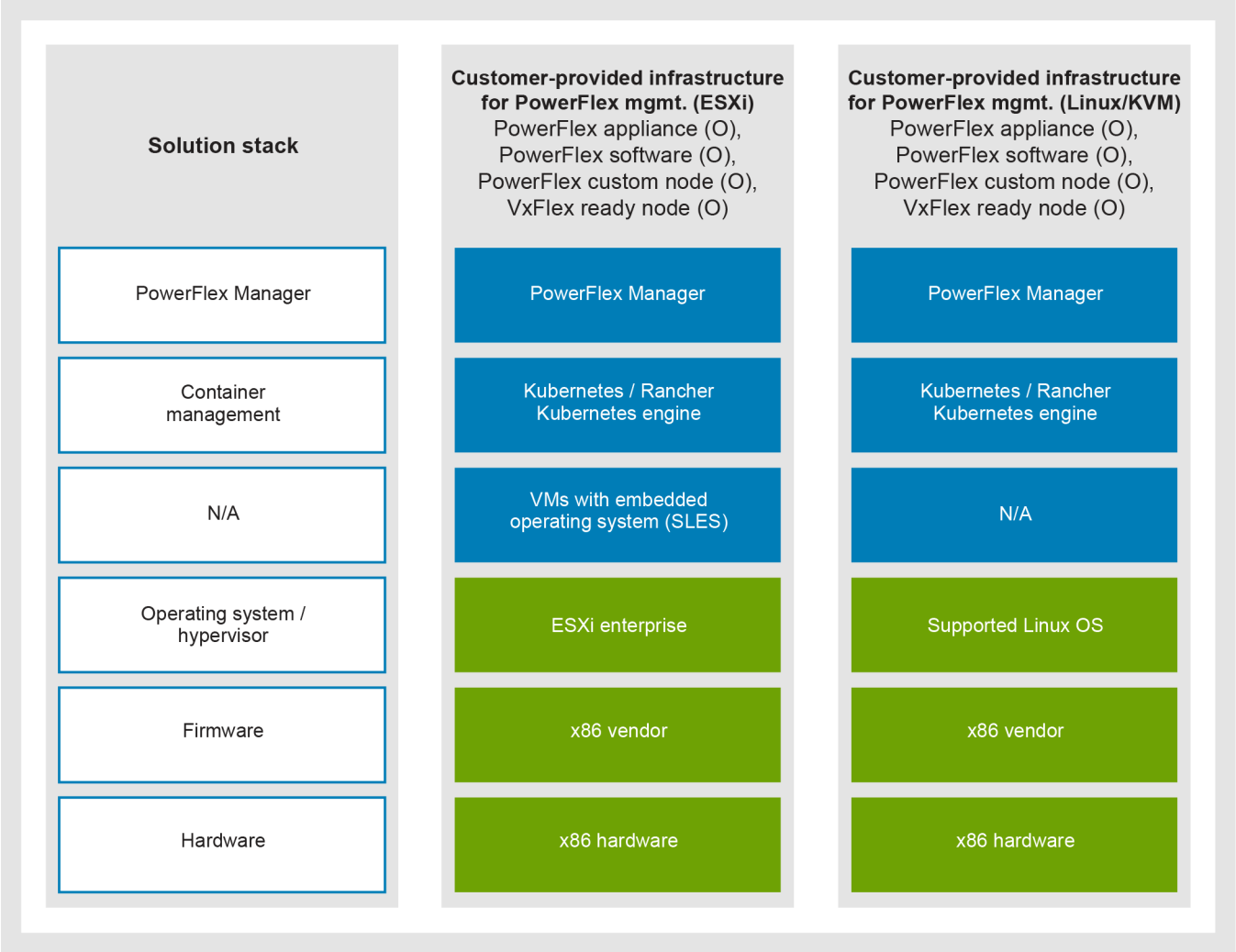


Figure 11. Customer-provided infrastructure for PowerFlex management model

8

Networking

In PowerFlex, inter-node communications manage data locations, rebuild and rebalance, and access applications to stored data on one IP network, or on separate IP networks.

Use any of the management interfaces to perform management tasks in the following ways:

- Through a separate network with access to the other PowerFlex components
- On the same network

You can configure these options during deployment using the CSV topology file or a PowerFlex Manager template for PowerFlex appliance or PowerFlex rack, or after deployment using the SCLI (PowerFlex CLI).

Related information

Installing the MDM in multiple-rack deployments

Network requirements and limitations

The following table includes the list of PowerFlex network requirements:

The following table includes the list of PowerFlex network requirements and supported architectures or configurations.

Table 25. Network requirements

Item	Requirement
Network type	Ethernet based
Minimum supported speed	2*10 GbE
Recommended speed	2*25 GbE
Certified configurations	4*25 GbE, 2*100 GbE, 4*100 GbE
Supported MTU	1500/9000 (jumbo frames recommended for optimal performance)
High availability and load-balancing	At least two physical ports with: ● PowerFlex native multipath across two or more subsets. (Up to eight subnets supported.) ● Link aggregation. A single IP subnet can be used with link aggregation; otherwise a minimum of two subnets are required. Dell Technologies recommends that you use MDM virtual IPs for simplified network stability in case of failures.
IP addresses	For more information about IP address calculation see *PowerFlex Custom Node Hardware Configuration and Operating System Installation Guide*.
Data network latency requirements for MDMs	Maximum of 300 milliseconds
Switch redundancy	● Switches must be redundant, providing continued access to components inside the rack in the network in case an access switch fails. ● Flat network is supported. ● Leaf-spine architecture is recommended for cross-rack traffic. ● VXLAN is supported.

Table 25. Network requirements (continued)

Item	Requirement
SDC connectivity	Connection through a router if supported and required for a low-latency connection for optimal performance.
PowerFlex single management subnet	Optional dedicated port or a VLAN on a trunk connection. For best practices, see *PowerFlex Appliance Network Planning Guide*.

Network limitations

The following table describes the PowerFlex network limitations:

Table 26. Network limitations

Item	Limitation
RMDA	Not supported
InfiniBand	Not supported
IPv4	Supported on all PowerFlex offerings.
IPv6	Supported only on PowerFlex software offerings.
DHCP	Cannot be deployed on a network in which PowerFlex MDMs, storage data servers (SDSs), storage data replicators (SDRs), or storage data clients reside.
Firewall rules.	Must allow PowerFlex ports. For port information, see the *Dell PowerFlex Family 4.x Security Configuration Guide*.

Network requirements for replication

The following table describes the additional network considerations and limitations for PowerFlex systems with asynchronous replication.

Table 27. Network requirements and considerations for systems with asynchronous replication

Item	Requirement or consideration
Recommended minimum network speed	4*25 GbE or 2*100 GbE
Additional IP addresses for site-to-site routing	• In the backend storage network, each SDR listens on the same node IP addresses as the SDSs and must be able to reach all SDSs within the protection domain. **NOTE:** Within a protection domain, SDRs are installed on the same hosts as SDSs. The traffic that an SDR writes to the journal volume is sent to all SDSs that host the journal, not only the one that is co-located on the host. • The SDRs require additional, distinct IP addresses that allow them to communicate with remote SDRs. Usually, these should be routable addresses with a properly configured gateway. For redundancy, each SDR should have two.
Static routes	• PowerFlex asynchronous replication usually occurs over a WAN between physically remote clusters that do not

Table 27. Network requirements and considerations for systems with asynchronous replication (continued)

Item	Requirement or consideration
	share the same address segments. If the default route is not suitable to properly direct packets to the remote SDR IP addresses, static routes must be configured to indicate either the next hop address or the egress interface, or both, for reaching the remote subnet. • PowerFlex Manager can add static routes for replication use cases.

The following table describes the network limitations for PowerFlex systems with asynchronous replication:

Table 28. Network limitations for systems with replication

Item	Limitation
MDM to MDM peer metadata synchronization	Should take place over a WAN with less than 200 msec latency.
Local SDR-to-remote SDR traffic	Latency is not as sensitive in SDR greater than SDR traffic, but round-trip time should not be greater than 200 msec.
Firewall rules	Must allow PowerFlex ports. For port information, see the *Dell PowerFlex Family Security Configuration Guide* for your PowerFlex deployment.

Network considerations

The following table explains the supported PowerFlex network configurations and the benefits of each of them. Select the network type for your deployment based on the requirements of your organization, security considerations, performance needs, and IT environment.

Table 29. Network configurations

Network type	Benefits
Single IP network All communications and I/Os used for management and for data storage are performed on the same IP network.	• Ease of use • Fewer IP addresses required
Multiple separate IP networks Separate networks are used for management and for data storage, or separate networks are used within the data storage part of the system.	• Security • Redundancy • Performance • Separate IP roles to enable separation between customer data and internal management

Additional considerations:

- Network high availability can be implemented using either NIC-bonding or several data networks in PowerFlex.
- The MDM cluster IP address limit is 16 IP addresses, which include all cluster members (primary, secondary, tiebreaker, and standby).
- For more information regarding MTU performance considerations and best practices, see the *PowerFlex and PowerFlex Manager Software Install and Upgrade Guide*.

IP address configurations

The following table describes the range of potential IP address configurations for PowerFlex:

Table 30. IP addresses for PowerFlex

Address type	Purpose	Description
MDM Management IP address	Management access	• Optional, but recommended • This is not applicable for tiebreaker IP addresses that can be used to provide access to PowerFlex management applications, such as PowerFlex Manager, CLI, REST API, and OpenStack. • This IP address must be externally accessible.
MDM IP addresses	Network control	• Mandatory IP addresses used for MDM control communications with SDSs and SDCs. Used to convey data migration decisions, but no user data passes through the MDM. • Must be on the same network as the data network. • Must be externally accessible if no MDM Management IP addresses are used.
SDS all IP addresses	Rebuild and network data path	• IP addresses used for both SDS-SDS and SDS-SDC communications. • These IP addresses will also be used to communicate with the MDM.
SDS-SDS-only IP addresses	Network rebuild	• IP addresses used for SDS-SDS communication only. • These addresses are used for rebuild and rebalance operations. • These IP addresses will also be used to communicate with the MDM.
SDS-SDC-only IP addresses	Network data path	• IP addresses used for SDS-SDC communication. • These addresses are only used for read/write user data operations.

The following combinations can be used for SDS or SDC:

- Only *SDS All IPs*
- Only *SDS-SDS Only IPs* + *SDS-SDC Only IPs*
- *SDS All IPs* + either *SDS-SDS Only IPs* or *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).
- *SDS All IPs* + both *SDS-SDS Only IPs* and *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).

NOTE: On Linux nodes, only the MDM needs a management IP address.

NOTE: On VMware, all Storage VMs (SVMs) need to have a management IP address and another address for the data network, the network on which traffic flows between SDSs and SDCs for read/writes, rebuild, and rebalance.

Network limitations

The same IP subnet can be configured on different VM kernel interfaces and virtual switches when deploying PowerFlex on VMware ESXi.

For more information, see Multihoming on ESXi.

PowerFlex only supports the following network configurations when deploying on VMware ESXi:

- A single data storage network
- Two or more data networks, each on separate IP subnets

- A single IP address data network using several NIC-bonding configurations, or vSwitch load-balancing

Network stability

Resilient network connectivity is standard for PowerFlex. Nevertheless the actual implementation may vary. In some cases, resiliency is attained through LACP or bonding, while in other cases through application-level multipathing.

PowerFlex has additional capabilities to support partial failures on existing connections. While many network issues are manifested as a complete disconnection, some are not. PowerFlex is aware of degraded connections, including:

- Stable, but slow connections in relation to other connections.
- Unstable connections (occasional disconnections).
- A combination of stable and unstable connections.

Additional PowerFlex capabilities support the following error scenarios:

- PowerFlex contains safeguards to maintain the proxy connection between the SDC and SDS. In the PowerFlex architecture, each SDC is required to be fully connected to all the SDSs in each of the protection domains from which it consumes volumes. In cases where the SDC is unable to reach one or more of the SDSs, the MDM reflects that and the management API raises an alert. The SDC then sends the I/O to another SDS within the same protection domain, or subset of this protection domain. The new SDS is referred to as a proxy SDS, because its only role is to forward the I/O to the SDS which owns the data, receive the I/O result, and return the result to the initiating SDC.
- PowerFlex supports using the MDM secondary IP addresses as a proxy connection to the primary MDM. This capability is applicable only when MDM VIP is not in use, because it requires configuration of all the SDC IP addresses to attempt a proxy connection.

9

Implementing PowerFlex

PowerFlex is based on a configuration of hardware and software components working together.

The PowerFlex physical layer consists of the hardware (servers with storage devices and the network between them) and the PowerFlex software that is installed on them. Communications are carried out over the existing LAN data network using standard TCP over IP. The MDM and SDS nodes can be assigned to up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

Implementing PowerFlex is, in general, a two-step process: first build the physical storage layer, then configure the virtual SAN layer on top of it.

To build the physical storage layer:

1. Physically assemble the server nodes in the rack and perform all cabling, according to the instructions for your PowerFlex product.
2. Configure the operating system, including all required IP addresses, based on network best practice guidelines.
3. Using PowerFlex Manager, deploy the PowerFlex software.

The physical layer is now ready, and you can expose a virtual storage layer.

Hardware

Hardware can be the existing application servers used by the data center, or a new set of nodes (if, for example, you want to dedicate all nodes solely for running the PowerFlex SAN storage system).

The PowerFlex rack architecture includes a full data center solution including: rack, switches, PowerFlex server nodes, power, and cabling. Built for capacity, performance, or compute per the specific configuration.

- Servers are the basic computer units used to install and run PowerFlex. They can be the same servers used for the applications (server convergence), or a dedicated cluster. In any case, PowerFlex is hardware-agnostic so aside from performance considerations, the type of server is inconsequential.
- The storage media can be any storage media, in terms of the type (SSD or NVMe SSD) and anywhere (external).

Support for Dell PowerEdge R660, R760, R860, R6625, and R7625 servers

PowerFlex certifies support of the following platforms for all offerings.

- Dell PowerEdge R660, Dell PowerEdge R760, and R860 servers (Intel based).
  - Available with a persistent memory solution, SDPM.
- Dell PowerEdge R6625 and Dell PowerEdge R7625 servers (AMD based) on PowerFlex compute - only nodes.

Physical layer

To implement the physical layer, perform the following steps:

1. Install the MDM component on the MDM nodes in one of the following configurations:
   - Three-node redundant cluster (one primary MDM, one secondary MDM, and one tiebreaker).
   - Five-node redundant cluster (one primary MDM, two secondary MDMs, and two tiebreakers).
   - Single node (one primary MDM).

   NOTE: It is not recommended to use single mode in PowerFlex systems, except in temporary situations. The MDM contains all the metadata that is required for system operation. Single Mode has no protection, and exposes the system to a single point of failure.

MDMs do not require dedicated nodes. They can be installed on nodes hosting other PowerFlex components.

2. Install the SDS component on all nodes that contributes some, or all, of their physical storage. Typically, each SDS is physically on a separate server.
   - Divide the SDS nodes into protection domains. Each SDS can be a member of only one protection domain.
   - Per protection domain, divide the physical storage units into storage pools, and optionally, into fault sets.
3. Install the SDC component on all nodes on which the application accesses the data exposed by the PowerFlex volumes.
   - In the following figure, the gray boxes represent a storage node which hosts the SDS, SDC, and MDM where applicable. In the cases where the gray boxes host the SDS, SDC, and MDM it is a three-node MDM cluster.



Figure 12. Physical layout example—three-node MDM cluster

Communication is done over the existing LAN using standard TCP over IP. The MDM and SDS nodes can be assigned to up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

SAN virtualization layer

After you install and configure the storage data client (SDC) components, expose the virtual SAN devices to the servers on which the SDCs are installed.

The MDM cluster manages the entire system. It aggregates the entire storage that is exposed to it by all the SDSs to generate a virtual layer - virtual SAN storage. Volumes can now be defined over the storage pools and can be exposed to the applications as a local storage device using the SDCs.

To expose the virtual SAN devices to your servers (the ones on which you installed and configured SDCs), perform the following:

1. Define volumes. Each volume that is defined over a storage pool is evenly distributed over all members using a RAID protection scheme. By having all SDS members of the storage pool participate, PowerFlex ensures:
   - Highest and most stable and consistent performance possible
   - Rapid recovery and redistribution of data
   - Massive IOPS and throughput

   You can define volumes as follows:

   - Thick

     Capacity is allocated immediately, even if not actually used. This can cause capacity to be allocated, but never used, leading to wasted capacity.

     Thick capacity provisioning is limited to available capacity.
   - Thin

     Capacity is "on reserve," but not allocated until actually used. This policy enables more flexibility in provisioning.

     Whereas thick capacity is limited to available capacity, thin capacity provisioning can be oversubscribed, as follows:

     Maximum thin capacity provisioning = 5 * (total raw capacity - used capacity)

     When capacity usage reaches the level where it may cause I/O errors, alerts are generated. At certain higher capacity levels, volumes (even thin volumes) can no longer be created.

     Example:

     In a system with 3 SDSs, each with 10 TB, there are 30 TB of storage.

     In the system, there is already a thick-provisioned volume that takes up 15 TB of the total raw capacity (created by adding a 7.5 TB volume).

     MDM will allow a total of 300 TB total raw capacity to be provisioned, and since 15 TB are already allocated, you can add a thin-provisioned volume of 285 TB total raw capacity (by adding a 142.5 TB volume) or a thick-provisioned volume of 15 TB total raw capacity.

     NOTE: This example uses 10x fine granularity over provisioning plus compression support.

2. Map volumes. Designate which SDCs can access the given volumes. This gives rise to the following:
   - Access control per volume exposed
   - Shared nothing or shared everything volumes

     Once an SDC is mapped to a volume, it immediately gets access to the volume and exposes it locally to the applications as a standard block device. These block devices appear as `/dev/scini`*X* where
     X is a letter, starting from "a."

     For example:

     - `/dev/scinia`
     - `/dev/scinib`
   - When a volume is defined on an AIX SDC, one device is created with the following pathnames:
     - A block device, named `/dev/scinid`*X*`...n`, where *X* is a number, starting from "0."
     - A raw device, named `/dev/rscinidX...n`, where *X* is a number, starting from "0."

     In general, mapping SDCs to AIX raw devices will yield best performance. If you are using the device to create a filesystem, use the block device.
   - The maximum amount of partitions for the scini disk is 15.
3. The maximum amount of volumes that can be mapped to an SDC is listed in *Product limits*.

   In a Windows environment, the device looks like any other local disk device, as shown in the Device Manager.

   NOTE: SDC mapping is similar to LUN mapping, in the sense that it only allows volume access to clients that were explicitly mapped to the volume.

Related information

Snapshots
Product limits

Audit logging

PowerFlex enables the redirection of the PowerFlex events and Ingress audit messages to syslog.

These events and audit messages are forwarded to an external Security Information and Event Manager (SIEM). You must perform the following configuration steps to enable this feature in PowerFlex Manager:

- Change Ingress settings to emit audit messages.
- Define a notification policy to forward Ingress audit messages in the PowerFlex system to an SIEM.
- Define a notification policy to forward events in the PowerFlex system to an SIEM.

For more information about the configuration steps, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.

Supported events

Audit logging supports the following event categories that occur in PowerFlex Manager and integrated components and systems:

- User authentication and authorization
- Changes to system configuration and security policies
- Security events

Naming PowerFlex objects

Provide meaningful names to all PowerFlex objects to make it easier to perform actions such as defining volumes and associating them with applications.

Storage pools can be named *Capacity_Storage* and *Performance_Storage*, which allows you to identify the different tiers.

For protection domains, one example would be separating the SDSs used by the finance department from SDSs used by the engineering department. This separation provides several benefits, including enhanced security. Thus, you might name the domains *Financial-PD* and *Engineering-PD*.

The fault sets could be called *FS_Rack01* and *FS_Rack02*.

Dell PowerFlex with Nutanix Cloud Platform

PowerFlex with Nutanix Cloud Platform is powered by Acropolis Hypervisor (AHV) for compute nodes and PowerFlex for storage nodes.

Nutanix Cloud Platform is traditionally a hyperconverged platform that consolidates computing, storage, and networking into a unified pool. It uses the Acropolis Operating System (AOS) for storage, AHV for virtualization, and Prism Central for centralized management across environments.

Nutanix Cloud Infrastructure Compute (NCI-C) consists of compute, networking, disaster recovery, and security.

The integration of PowerFlex with Nutanix Cloud Platform enables organizations to scale compute and storage resources independently. Nutanix AHV hypervisor and AOS operate on PowerFlex compute nodes, using storage from an external PowerFlex storage cluster in a two-layer configuration. This setup leverages PowerFlex software-defined storage alongside the Nutanix Cloud Platform infrastructure to optimize resource management. The architecture supports data protection, disaster recovery, and networking while facilitating hybrid cloud operations for scalable workloads.

Architecture

PowerFlex with Nutanix Cloud Platform is deployed using a two-layer architecture, as shown in the following figure.

- Compute layer: An NCI-C cluster with AOS and AHV running on PowerFlex compute-only nodes.
- Storage layer: A PowerFlex cluster operates in a storage-only model, compatible with any PowerFlex configuration. This cluster provides network-attached storage.

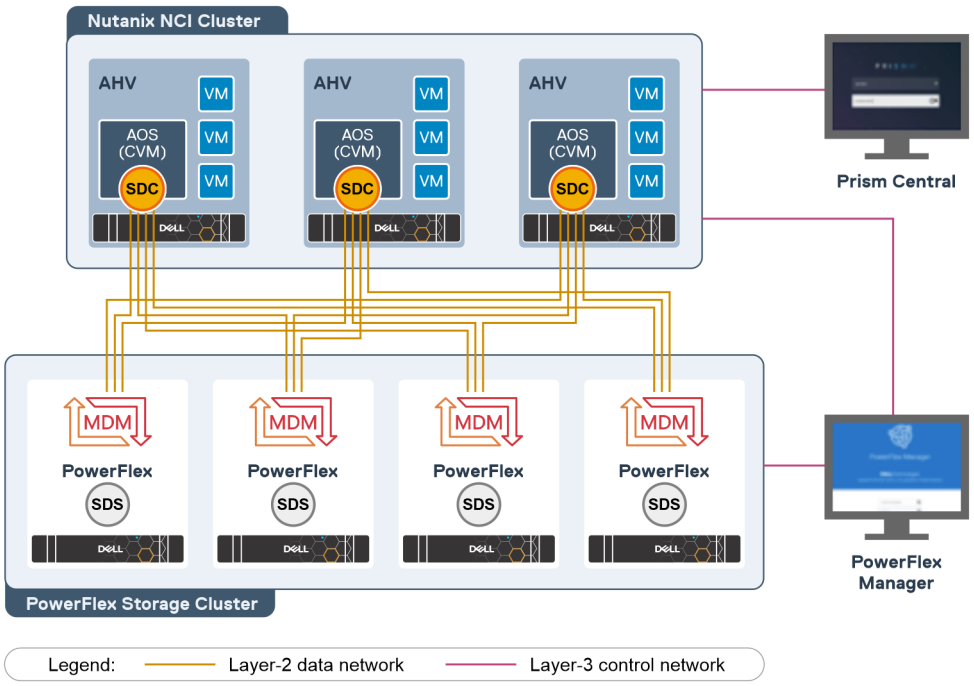


Figure 13. PowerFlex with Nutanix Cloud Platform architecture

Unlike the hyperconverged infrastructure (HCI) model, where storage is sourced from locally attached internal drives, the PowerFlex with Nutanix Cloud Platform solution architecture decouples the locally attached internal storage. In this setup, the compute cluster accesses storage over the network from the external PowerFlex storage cluster by the SDC that is installed on every node of the NCI-C cluster.

When you deploy PowerFlex with Nutanix Cloud Platform, you host NCI-C on the PowerFlex compute cluster, while you provision storage from the PowerFlex storage cluster.

PowerFlex implementation in a VMware ESXi-based system

In the VMware ESXi-based environment, the MDM, and SDS components are installed on a dedicated storage virtual machine, and the SDC is installed directly on the VMware ESXi host.

Minimum requirements

These are the minimum system requirements for a PowerFlex implementation in a VMware ESXi-based system:

- Three VMware ESXi servers with 240 GB of free capacity per server
- 10 Gbps network

Table 31. Minimum system requirements

Configuration type	Supported VMware ESXi versions	Supported PowerFlex offering
VMware ESXi-based hyperconverged nodes with SDC	ESXi 8.0	PowerFlex appliance PowerFlex rack

Table 31. Minimum system requirements (continued)

Configuration type	Supported VMware ESXi versions	Supported PowerFlex offering
NOTE: Compute nodes may also consume storage from VMware ESXi-based hyperconverged nodes.	Dell Technologies recommends implementing the latest update.	
VMware ESXi compute nodes with SDC connecting to a PowerFlex storage-only system	ESXi 8.0 Dell Technologies recommends implementing the latest update.	PowerFlex appliance PowerFlex rack PowerFlex software PowerFlex custom node VxFlex Ready Node
VMware ESXi compute nodes with NVMe over TCP connecting to a PowerFlex storage-only system	ESXi 8.0 Dell Technologies recommends implementing the latest update.	PowerFlex appliance PowerFlex rack PowerFlex software PowerFlex custom node VxFlex Ready Node

NOTE: For a full list of supported operating systems, see the PowerFlex support matrix.

Components

The PowerFlex hyperconverged configuration consists of a dedicated storage virtual machine and these software components:

- Storage virtual machine: A Linux-based virtual machine that is dedicated to PowerFlex and is used to host the different PowerFlex software components described here.
- Meta data manager (MDM): Configures and monitors the PowerFlex system. The MDM is configured in a redundant cluster mode. The MDM is installed on the storage virtual machine. The MDM can be configured as a three-node cluster (primary MDM, secondary MDM, and tiebreaker MDM) or as a five-node cluster (primary MDM, two secondary MDMs, and two tiebreaker MDMs) to provide greater resiliency.
- Storage data server (SDS): Manages the capacity of a single server and acts as a backend for data access. The SDS is installed on all servers contributing storage devices to the PowerFlex system. These devices are accessed using SDS. The SDS is installed on the storage virtual machine.
- Storage data replicator (SDR): Handles volumes replication activities between PowerFlex systems through journal management. The SDR is located alongside the SDS.
- Storage data client (SDC): A lightweight device driver that presents PowerFlex volumes as block devices to the application on the server which the SDC is installed. The SDC creates a logical adapter, which is a VMware ESXi kernel construct. The adapter informs VMware ESXi about the arrival and disappearance of SCSI devices. These LUNs can be formatted with VMFS and then exposed using the VMware ESXi host to the virtual machine or can be used as RDM devices.

NOTE: NVMe target (SDT) is not supported in VMware ESXi-based hyperconverged configuration.

This implementation is shown in the figure. The SDC is installed within the VMware ESXi kernel similar to any other VIB.

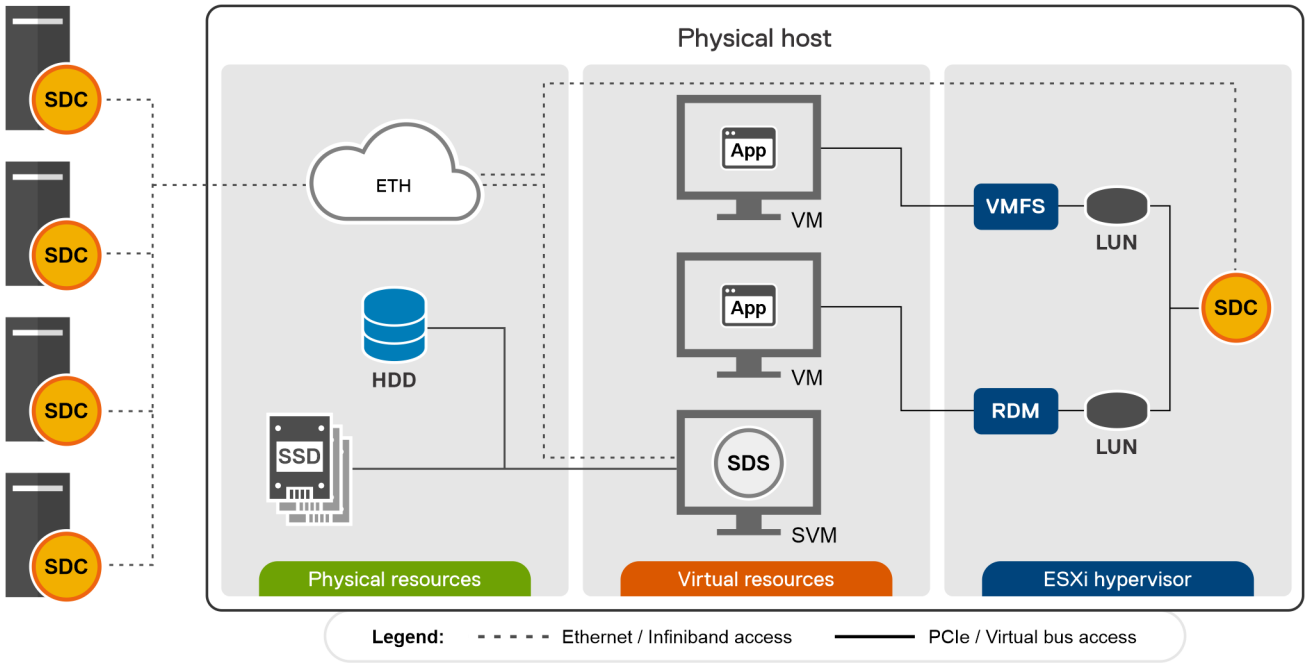


Figure 14. PowerFlex implementation in VMware ESXi with SDC in VMkernel

The LUNs in this example can be formatted with VMFS, and then exposed using the VMware ESXi host to the virtual machine, or the LUNs (volumes) that are NVMe target based can be used as RDM devices. When the LUNs are used as RDM devices, the VMFS layer is omitted.

Supported features

The following VMware ESXi features are supported by PowerFlex:

- vMotion
- Storage vMotion
- Fault tolerance
- DRS
- Storage DRS
- VAAI (except full copy primitive)

Device management

Devices can be managed with VMDirectPath I/O or RDM.

Device management with VMDirectPath I/O

Device management is performed by the storage virtual machine, yielding the best I/O performance. Devices are added to the system after the deployment. On factory-installed VxFlex Ready Node servers, the vSphere plug-in configures DirectPath on the ESXi servers and adds devices to the system. On other servers, you must enable VMDirectPath manually to the ESXi server, then use the plug-in to add devices.

- When DirectPath is configured, all servers in the system must use DirectPath. For PowerFlex systems, this can be configured with the PowerFlex vSphere plug-in. For other servers, use the vSphere client to configure each ESXi host manually.
- If the host has multiple controllers, you must configure DirectPath on that host manually, not with the plug-in. After that is done, you can use the plug-in to deploy PowerFlex on that host and to add devices to it.
- All devices on all servers must not have any other use (not VMDK, RDM, or be part of a datastore).
- Dell PowerEdge R840, Dell PowerEdge R740xd, and Dell PowerEdge R640 servers with any NVMe devices cannot be added in a DirectPath-based system. You must use RDM-based.

ESXi boot device requirements are as follows:

- Must be on a separate controller or connected directly to the board.
- SATADOM and M2 boot devices are supported. These devices allow the creation of a datastore on the system disk, which is required to host the storage virtual machine.
- USB boot devices are not supported.

NOTE: DirectPath architecture is the recommended best practice for RAID, and SAS Controller managed drives.

Device management with RDM

Using RDM mapping, a device is created on the storage virtual machine that points to the hard drive on the ESXi. You can add RDM devices that are connected through a physical RAID controller. If a local RDM is not connected by a RAID controller, it may not be supported.

Before enabling this feature, contact Dell Technologies Support.

System size

If you are deploying a very large PowerFlex system (hundreds of nodes), you can increase the default parallelism limit (default: 100), thus speeding up the deployment. This depends on the processing power of the vCenter.

Pre-deployment considerations

You should take these considerations into account before deploying the system:

- Whether separate networks will be used for data and management (recommended).
- The IP addresses that will be used for the storage virtual machines and VMkernels.
- If flash devices will be added to the SDS.
- If you want to create fault sets.

Post-deployment considerations

You should take these considerations into account after deploying the system:

- After deployment is complete, set all storage virtual machines to start automatically with the system. Do not set storage virtual machines under the VMware resource-pool feature.
- In a DirectPath environment, after deploying the system, you must add devices to the SDS.

Related information

Fault sets

VMware ESXi integrated feature support

This section describes VMware ESXi integrated feature support for PowerFlex.

VMware ESXi vStorage APIs for Array Integration

VMware ESXi vStorage APIs for Array Integration provides hardware acceleration functionality. It allows the host to offload specific virtual machine and storage management operations to compliant storage hardware. With the assistance of the storage hardware, the host performs these operations faster, and consumes less CPU, memory, and storage fabric bandwidth.

VAAI uses these fundamental operations:

- Atomic Test & Set, which is used during creation and locking of files on the VMFS volume.
- Clone Blocks/Full Copy/XCOPY, which is used to copy or migrate data within the same physical array.

- Zero Blocks/Write Same, which is used to zero-out disk regions.
- Thin Provisioning in VMware ESXi 7.x and later hosts, which allows the VMware ESXi host to tell the array when the space previously occupied by a virtual machine (whether it is deleted or migrated to another datastore) can be reclaimed on thin provisioned LUNs.
- Block Delete in VMware ESXi 7.x and later hosts, which allows for space to be reclaimed using the SCSI UNMAP feature.

The PowerFlex supported VAAI features are:

- Atomic Test & Set
- Zero Blocks/Write Same
- Thin Provisioning in VMware ESXi 7.x and later hosts
- Block Delete in VMware ESXi 7.x and later hosts

The following is an output example:

```
esxcli storage core device vaai status get -d
eui.7dbf14034834bbe01bf7e55800000002
eui.7dbf14034834bbe01bf7e55800000002
VAAI Plugin Name:
ATS Status: supported
```

`Clone Status: unsupported` This means that Clone Block/Full Copy/Xcopy is not supported.

`Zero Status: supported` This means that write same is supported.

`Delete Status: supported` This means that UNMAP is supported.

NOTE: Thin provisioning is not shown in VAAI output.

VMware vSphere Storage I/O Control

While VMware Sphere Storage I/O Control (SIOC), Storage I/O Statistics Collection (SIOSC), and Network I/O Control (NIOC) are useful for vSan environments, their implementation may cause significant issues in a PowerFlex environment. Therefore, use of these options are not supported.

PowerFlex provides integrated capabilities to limit network bandwidth and IOPS limits for each volume for each storage data client.

Supporting VMware ESXi

PowerFlex Manager supports VMware ESXi 8.0 and VMware vCenter 8.0. VMware vCenter 8.0 supports the deployment of VMware ESXi-based hyperconverged infrastructure or managed compute nodes with VMware ESXi 8.0 and service operation. VMware ESXi 8.0 support is equivalent to the current support for VMware ESXi 7.x.

VMware ESXi 8.0

Upgrade from VMware ESXi 7.x to VMware ESXi 8.x by performing a non-disruptive update operation on existing deployments. Install VMware ESXi 8.x on new deployments of PowerFlex.

Upload complete VMware ESXi software bundles

Upload the entire VMware ESXi software bundle in a ZIP format, from a local drive or a network location. On completion of the node discovery activity, the BIOS configuration is implemented using iDRAC.

VMware ESXi upgrade

PowerFlex Manager supports enabling the VMware ESXi upgrade and allows upload of external VMware ESXi upgrade images.

The following upgrades can be done using PowerFlex Manager for appliance and rack systems that are on 4.x:

- VMware ESXi 8.x to 8.x upgrades

- VMware ESXi 7.x to 8.x upgrades
- VMware ESXi 7.x to 7.x upgrades

Consumption models

PowerFlex is offered in the following consumption models: PowerFlex appliance, PowerFlex rack, PowerFlex software, PowerFlex custom node and VxFlex Ready Node, PowerFlex for AWS, and PowerFlex for Microsoft Azure.

PowerFlex software

The following table describes the components in the PowerFlex software offering:

Table 32. PowerFlex software components

Component	Software management catalog
PowerFlex software	Included
Operating system	Not included. See the *PowerFlex Support Matrix*.
System firmware • BIOS • Cards • Drives	Customer managed
Drivers	Customer managed
Infrastructure firmware • Cabinet/PDU • IPI Appliance	N/A
Dell CloudLink	See the *PowerFlex Support Matrix*
VMware NSX	Customer managed
Network switch firmware and operating system	Customer managed

PowerFlex appliance

PowerFlex appliance is a scalable system with flexible form factors that comes pre-configured and validated for fast, easy deployment.

Table 33. PowerFlex appliance components

Component	PowerFlex appliance Intelligent Catalog
PowerFlex software	Included
Operating system and drivers	Included
System firmware • BIOS • Cards • Drives	Included
Infrastructure firmware • Cabinet / PDU • IPI Appliance	N/A
CloudLink	Included
VMware NSX	Customer-managed

Table 33. PowerFlex appliance components (continued)

Component	PowerFlex appliance Intelligent Catalog
Network switch firmware	Full network automation is included Partial network automation: See the *PowerFlex Support Matrix*

PowerFlex rack

PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support.

Table 34. PowerFlex rack components

Component	PowerFlex rack Release Certificate Matrix
PowerFlex software	Included
Operating system and drivers	Included
System firmware • BIOS • Cards • Drives	Included
Infrastructure firmware • Cabinet/PDU • IPI Appliance	Included
CloudLink	Included
VMware NSX	Customer-managed
Network switch firmware	Included

PowerFlex custom node and VxFlex Ready Node

PowerFlex custom node is a do-it-yourself networking and management solution in which PowerFlex is installed on Dell PowerEdge servers that have been optimized for PowerFlex systems.

Table 35. PowerFlex custom node and VxFlex Ready Node components

Component	Software management catalog
PowerFlex custom node and VxFlex Ready Node	Included
Operating system	See the *PowerFlex Support Matrix*
System firmware and drivers • BIOS • Cards • Drives	See the *PowerFlex Support Matrix*
Infrastructure firmware • Cabinet/PDU • IPI Appliance	N/A
CloudLink	See the *PowerFlex Support Matrix*
VMware NSX	Customer-managed
Network switch firmware	Customer-managed

Dell PowerFlex for Public Cloud

Dell PowerFlex for Public Cloud is a deployment of Dell PowerFlex, software-defined block storage innovation, in the public cloud. PowerFlex for Public Cloud is available for both AWS and Microsoft Azure allowing you to experience the same benefits of enterprise-class storage services in the cloud as with on-premises. It provides higher performance, larger volume sizes, and improved resilience than what is currently available on the public cloud.

Table 36. Capabilities and benefits

Capability	Benefit or Value
Unique multi-availability zone durability	• Enhanced, space efficient data protection through the aggregation of all instances across availability zones for volume provisioning. • Enables high resilience across availability zones by protecting the data during availability zone failure, without replication of data (space and IOPs).
Extreme performance for mission-critical workloads running in the cloud	Meets and exceeds SLAs with extreme performance (high throughput and low latency) for workloads such as databases and analytics.
Scalable, flexible, and resilient	Meets stringent SLAs and run workloads with confidence and assurance with near linearly scalable performance as workload demand increases.
Data mobility	Seamless data mobility with the ability to easily move data from on-premises to the cloud or across regions within the cloud as workloads demand increases.
Self-healing architecture with rapid rebuild	Ensure high availability and service availability under failure conditions with fast reprotection and rebuild.
Robust ecosystem of automation tools and frameworks (CSM/CSI, Ansible, REST APIs, DDVE, CloudLink, Dell AIOps)	Optimal usage and performance of cloud-based block storage with easy to deploy, expand, monitor, and manage capabilities.

NOTE: The following features that are supported by PowerFlex are NOT supported with PowerFlex for Public Cloud: compression, fine granularity storage pools in the public cloud, and NVMe over TCP.

PowerFlex for AWS

PowerFlex for AWS empowers enterprises to run diverse workloads in the public cloud while ensuring extreme performance, scalability, and a simplified cloud experience. PowerFlex for AWS can be deployed in two configurations by using Elastic Block Store (EBS) volumes, or native NVMe SSD drives attached to EC2 instances (EC2 instance store). PowerFlex for AWS also provides the proven enterprise data services, such as thin provisioning, snapshots, and asynchronous replication, required to run demanding block-based workloads in the public cloud. PowerFlex for AWS native asynchronous replication enables data mobility between on-premises and the cloud or across regions in the cloud, for example, AWS East to AWS West.

PowerFlex for Microsoft Azure

Now available PowerFlex for Microsoft Azure, which can be deployed in two configurations. Managed Disks or virtual machines with attached NVMe SSDs based on the use case. PowerFlex for Microsoft Azure also provides the proven enterprise data services, such as thin provisioning, snapshots, and asynchronous replication, required to run demanding block-based workloads in the public cloud. PowerFlex for Microsoft Azure native asynchronous replication enables data mobility between on-premises and the cloud or across regions in the cloud.

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Maintenance

Maintenance of PowerFlex is primarily limited to configuration changes of the physical and virtual layers. It requires minimal user attention. When maintenance or planned restart of an SDS is required, the maintenance mode feature can be used to streamline system operation.

Maintain the physical layer

Adding or removing hardware units and configuring them into PowerFlex can result from scale-out, hardware failure, operating system updating, or upgrade and hardware expansion.

In the physical layer, maintenance is limited to adding and removing hardware units and configuring them into PowerFlex. These operations are usually a result of:

- Scaling out when there is a need for additional capacity. This usually results in adding more storage media to the existing servers, or adding additional servers.
- Hardware failure. In cases where there is a hardware (storage media or server) failure and it must be replaced.

  In all the above cases, the operation requires adding or removing storage capacity from the system. In some cases, it may include adding or removing an entire server, and its associated storage media, from the configuration. As far as PowerFlex is concerned, all these activities translate to SDS reconfigurations.

  If the removed server is an SDC node, or the server to be added requires exposing storage locally, SDC reconfiguration will happen as well.
- Adding or removing storage media. Add or remove the media from the SDS with which it is associated. PowerFlex will redistribute the data accordingly and seamlessly.
- Adding or removing a node. Add or remove the SDC and SDS residing on the node. PowerFlex will redistribute the data accordingly and seamlessly.
- Operating system updating, or upgrade or hardware expansions. Graceful shutdown or reboot operations with return to fully operation state flow.

Instant maintenance mode

Using instant maintenance mode, you can restart a server hosting an SDS without requiring a shutdown or interrupting application I/Os. Storing all writes created during maintenance to a dedicated fault set prevents data loss if there is a single failure.

Instant maintenance mode enables you to restart a server that hosts an SDS, with minimal impact on PowerFlex, thus bypassing the disruption and effort caused by disorderly shutdown, protection domain shutdown, and orderly shutdown.

Whereas PowerFlex always uses two copies of user data, invoking maintenance mode introduces an additional copy that stores all writes created during maintenance to an SDS or fault set (created during maintenance) in both a primary location and a new location. This copy prevents data loss if a single failure occurs.

When the SDS or fault set is returned from maintenance mode, only the new writes are required to be resynchronized, thus minimizing data transfer during and after the update.

Instant maintenance mode does not interrupt application I/Os; it can be run on any amount of members of a fault set; and it can run in parallel on different protection domains. While an SDS is in maintenance mode, most PowerFlex operations (like adding a volume) cannot be performed in the fault set, protection domain, or storage pool in which the SDS and its devices reside.

To invoke maintenance mode, the following conditions are required:

- Only one fault unit (or stand-alone SDS) can be in maintenance mode at any given time.
- No other SDSs can be in a degraded or failed state (force override can be used).

- There must be adequate space on other SDSs for the additional backup (force override can be used).

  NOTE: Use of force override options when entering maintenance mode can lead to data unavailability while maintenance mode is activated.

  While an SDS is in maintenance mode, it can be shut down with no danger to data.

Related information

Rebalance throttling

Protected maintenance mode

Protected maintenance mode is anther type of maintenance mode that ensures no data loss during down time.

Protected maintenance mode is another type of maintenance state where a third copy is created. If there should be a node failure during instant maintenance mode, then there is no SDS up to serve I/Os for the data. With protected maintenance mode, a third copy is created before maintenance mode, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS.

Protected maintenance mode includes the following:

- Spare capacity accounting and rebuild operations to accurately reflect ongoing capacity reclamation activities.
- Workflows with reclamation-aware completion handling, ensuring reclamation activities are completed before protected maintenance mode operations conclude.
- Capacity reclamation progress monitoring capabilities for fine granularity storage pools.
- Capacity reclamation and rebalance efficiency for fine granularity storage pools to optimize capacity recovery and management during rebuild and recovery operations.

NOTE: The maintenance mode that Dell Technologies recommends is protected maintenance mode. Use instant maintenance mode only if protected maintenance mode cannot be used.

Auto abort

Protected maintenance mode consumes a significant amount of capacity. In some cases, this capacity may be required for more important tasks such as user I/O or rebuild.

The purpose of this capability is to cancel protected maintenance mode when approaching the end of capacity, to avoid impacting I/O or rebuild. For example, if a single node is entering protected maintenance mode and a third copy is being created, then a second node fails. Auto abort will confirm that the third copy space is made available to perform a rebuild.

Maintain the virtualization layer

This section lists the operations that can be performed on volumes exposed by the PowerFlex virtual SAN.

The following operations may be performed on volumes that are exposed by the PowerFlex virtual SAN:

- Add or remove a volume:

  Create or delete a volume in the system.

- Increase volume size:

  Add capacity to a given volume, as needed. The change in volume size occurs seamlessly without interrupting I/O.

- Map and unmap volumes to an SDC:

  This enables or disables access to a volume by an SDC, and thus by an application residing on the same node.

- MDM operations:
  - Move the MDM role to another role
  - Add MDM VIP
  - Upgrade of MDM, SDS, SDC and LIA

Configure direct attached storage

When adding direct attached storage to SDS, it is recommended to configure the raw devices as stand-alone devices. Upon adding a device to an SDS PowerFlex verifies that the device is clear and returns an error if it is otherwise.

PowerFlex works with any free capacity—internal or direct-attached devices, either magnetic hard disk drives or flash-based devices such as solid state drive (SSD) and PCIe cards. Although PowerFlex can work with any device topology, it is recommended to configure the raw devices as stand-alone devices.

Device data is erased when devices are added to SDS. When adding a device to an SDS, PowerFlex checks that the device is clear before adding it. An error is returned, per device, if it is found not to be clear. You can override this check by using the force device takeover option.

The following devices are considered to be not "clear," and thus cannot be added to SDS:

- Linux - A complete device with either a file system or partition, or a partitioned device with a file system.
  - Multipath devices cannot be added as SDS devices.
  - Devices in an LVM group cannot be added to an SDS.
- VMware ESXi - Same as above, depending on the operating system of the SVM where the SDS is installed.

Limitations:

- SAN devices will not be prevented from being added.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

NOTE:

If the server has a RAID controller, PowerFlex prefers to use the caching abilities of the controller for better performance, but is better used when all devices are configured as stand-alone (that is setting each of the devices to RAID 0 separately). Dell Technologies recommends enabling RAID controller caching for hard disk drive devices. Caching is not needed for SSD devices.

NOTE:

For hard disk drives: It is recommended to use RAID controller caching when available as follows:

- READ/WRITE: If the cache is battery-backed
- READ ONLY: If the cache is NOT battery-backed

For SSDs (such as Flash): The setting should be write-through.

Monitoring

PowerFlex supports a number of software and hardware monitoring features to assist you with system maintenance.

SDC connection monitoring

The system monitors all connections between SDCs and SDSs and sends out an alert when an active connection between an SDC and an SDS goes down.

To monitor SDC and SDS connections effectively, the MDM collects connectivity updates from all of the SDCs. The MDM posts events whenever an SDC connects to or disconnects from a specific SDS IP address. The MDM frequently analyzes the connectivity status to determine the current system state. The system does not send out alerts for temporary connectivity issues that are resolved in less than 10 seconds.

The following are the possible connectivity states between SDCs and SDSs in the system:

- All connected
- One SDC is disconnected from one SDS
- One SDC is disconnected from one SDS IP address
- One SDC is disconnected from all SDSs
- All SDCs are disconnected from one SDS
- All SDCs are disconnected from one SDS IP address

- All SDCs are disconnected from all SDSs
- Multiple disconnections

When the connectivity state of the system changes to any state other than `All Connected`, an alert is displayed in PowerFlex Manager and is written to the MDM event log. Once an alert is generated, you can use the SCLI to query details on the disconnection using the command `scli --query_sdc_to_sds_disconnections`.

The MDM does not monitor the connectivity state of SDCs or SDSs in the following scenarios:

- SDS is in maintenance mode
- SDS is disconnected from the MDM
- SDS is in the process of being removed
- SDC is disconnected from the MDM for more than two minutes
- SDC is not approved

SMART hardware monitoring

The PowerFlex bare-metal solution provides monitoring capabilities for RAID controllers and storage devices compatible with SMART (Self-Monitoring, Analysis and Reporting Technology) protocols.

In Linux-based environments, SMART-compatible HDDs, SSDs and RAID storage controllers can be monitored for SMART attributes such as temperature, SSD wear level, and error counters. LEDs can also be lit on these hardware devices, to simplify physical identification for maintenance purposes.

Each hardware vendor defines specific thresholds for the SMART attributes. This feature currently supports storage devices controlled by LSI, HP and Dell RAID controllers, and stand-alone devices. During system deployment, an external monitoring tool is installed as part of the LIA on each node. Additional RAID controller tools must be installed manually after system deployment: storcli for LSI RAID controllers, hpssacli for HP RAID controllers, or perccli for Dell RAID controllers. These tools are used by the system to collect the counters that are returned to the MDM.

NOTE: In some cases, LSI RAID controllers may report vendor information as "AVAGO" instead of LSI.

The MDM queries the SDSs at set intervals, and stores the returned information. This information can be viewed using CLI queries. In addition, when thresholds are crossed for SMART attributes, alerts are generated by the system.

When the CLI is used to query device information, physical device information, such as serial number, disk slot information, model name, vendor etc., temperature, and wear level information (for SSDs only) is included in the returned response.

You can use PowerFlex Manager to monitor SMART-related alerts in the **Alerts** view.

In addition, SNMP traps and Secure Connect Gateway alert codes can be used to monitor alerts triggered by devices compatible with SMART.

List of approved RAID controllers

High-level specifications of RAID controllers, which are tested and certified by PowerFlex.

Table 37. PowerFlex-certified RAID controllers

Manufacturer	Specifications
Dell	• Model Name: PERC H730 Mini • Firmware Version: 25.5.9.0001 • Driver Version: 07.700.52.00 and above • Driver Name: megaraid_sas
	• Model Name: H755 • Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix. • Driver Version: See VxFlex Ready Node Firmware and Driver support matrix. • Driver Name: megaraid_sas
	• Model Name: H740p • Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix.

Table 37. PowerFlex-certified RAID controllers (continued)

Manufacturer	Specifications
	• Driver Version: See VxFlex Ready Node Firmware and Driver support matrix. • Driver Name: megaraid_sas
	• Model Name: HBA330 • Firmware Version: See VxFlex Ready Node Firmware and Driver support matrix. • Driver Version: See VxFlex Ready Node Firmware and Driver support matrix. • Driver Name: mpt3sas
HP	• Model Name: Smart Array P440ar • Firmware Version: 3.56 • Driver Version: 3.4.10 • Driver Name: hpsa

NVDIMM hardware awareness feature

From PowerFlex, the customer can enable the NVDIMM hardware awareness feature on the SDS.

When this feature is enabled, the SDS detects bad blocks on NVDIMM devices used for fine granularity acceleration and takes appropriate actions based on how and when the corruptions in the NVDIMM devices are detected.

Management diagnostics

Management diagnostics provide detailed information about the health of PowerFlex management platform pods, nodes, and diagnostics. Only superusers and support users have access to troubleshoot, monitor, and review management diagnostics.

The PowerFlex management platform **Diagnostics** page provides an overview of the latest scheduled run details and manually run diagnostic results.

The **Nodes** page displays the overall health of all the PowerFlex management platform nodes.

The **Pods** page displays the overall health of all the PowerFlex management platform pods. You can restart specified pods from the user interface, copy and download logs from the current pod life cycle or download historical logs from all pod life cycles.